导论

课程链接：
https://www.shenlanxueyuan.com/course/385

什么是空中机器人?

空中机器人 ≠ 军用无人机

- 尺寸大多很大
- 10+小时续航
- 远程控制
  - 指点飞行
  - 遥控器控制
  - 4~10位操控员

"Drones mischaracterize what these things are. They're not dumb. Nor are they unmanned, actually. They're remotely piloted aircraft."

-- Gen. Norton Schwarz, August 10, 2012

无人飞行器的发展史

第一架无人驾驶飞机的尝试

第一架可复用无人驾驶飞行器

复仇武器1号

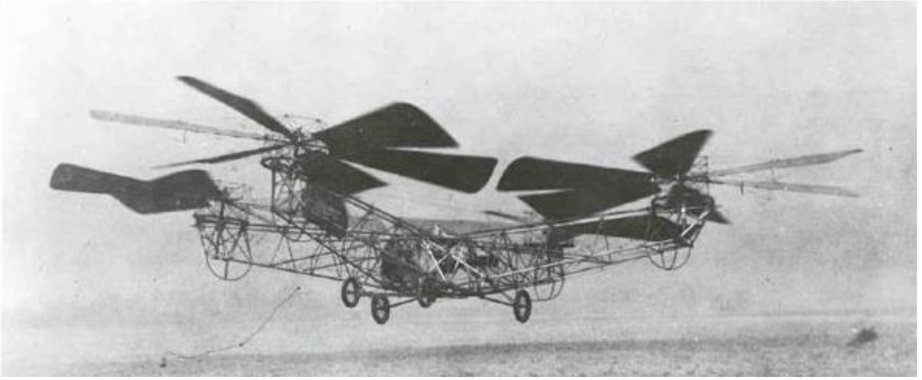
MQ捕食者无人机

全球鹰项目

民用无人机

多旋翼的发展史

- De Bothezat 直升机（1922）
  - 飞行高度5m
  - 完全靠手动控制的非自稳系统
- 接下来几十年无事发生
  - 几乎不可能的手动控制
  - 多旋翼难以大型化
  - 没有合适的传感器（传统IMU很大并且昂贵）
- STARMAC 斯坦福大学（2004）
  - 得益于智能移动设备的发展
  - MEMS传感器 + 嵌入式机载电脑

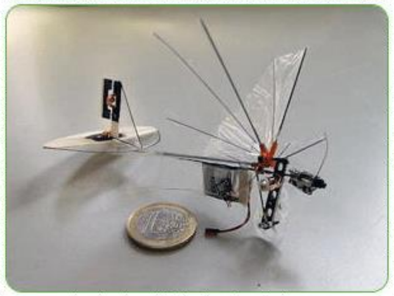
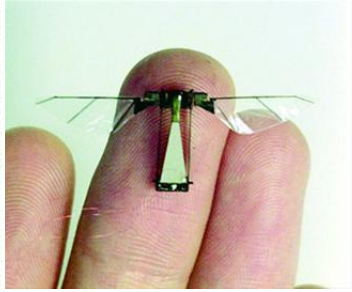


什么是空中机器人?

无人飞行器





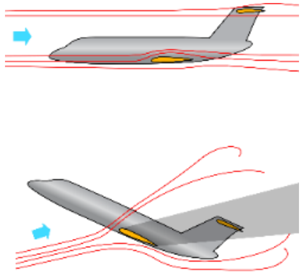




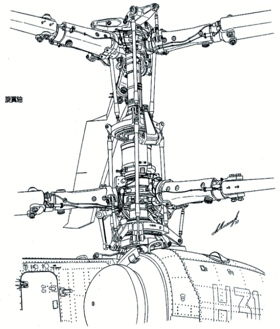
什么是空中机器人？

无人飞行器

无人飞行器（UAV）的种类

- 固定翼
  - 长续航，大负载，自稳系统 ☺
  - 需要跑道，可能失速，需要空气动力学设计 ☹
- 直升机
  - 垂直起降 ☺
  - 中等续航与负载
  - 复杂的机械设计，非自稳系统 ☹
- 多旋翼
  - 垂直起降，简易的机械设计 ☺
  - 难以大型化
  - 较短的续航和较小的负载，非自稳系统 ☹

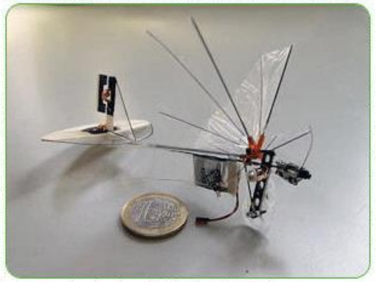


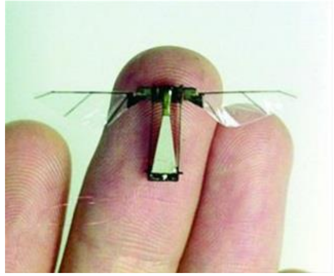


无人飞行器

更棒的设计?

- 混合式无人机（旋翼+固定翼）
    - 垂直起降，长续航，大负载 ☺
    - 技术未发展成熟 ☹
- 扑翼/仿生无人机
    - 适合小型系统
    - 技术未发展成熟





多旋翼小型无人飞行器

- 小尺寸（直接小于1米）
- 足够的负载（1~5kg）
- 低廉的价格（小于5万人民币）
- 安全
- 优越的机动性


侦查


搜索救援


运输


航拍

执法

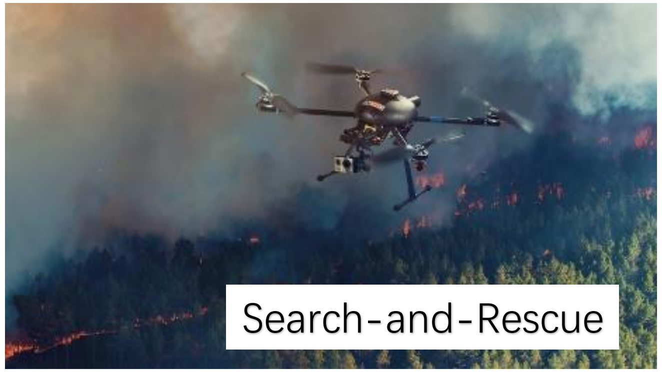

农业

Multirotor


Agriculture


Aerial mapping

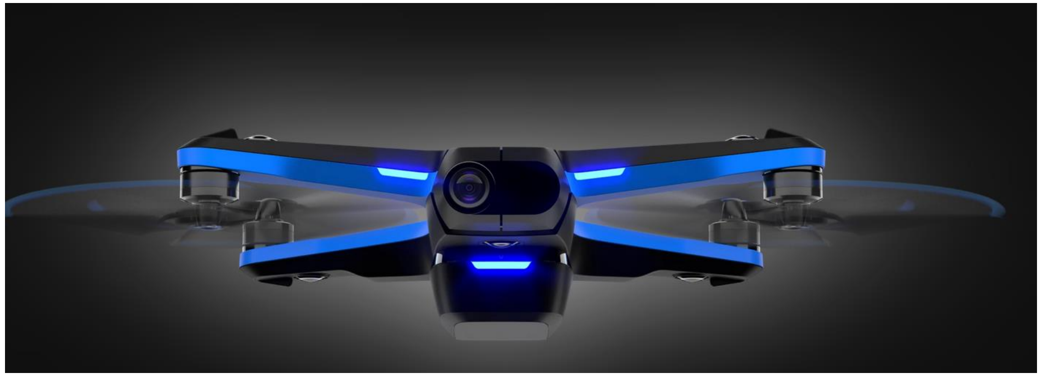

Aerial Filming


Search-and-Rescue

消费级无人机

大疆“晓”Spark

大疆 Mavic Air 2S



Skydio 2

穿越机

自组穿越机

DJI FPV

行业应用

农业植保

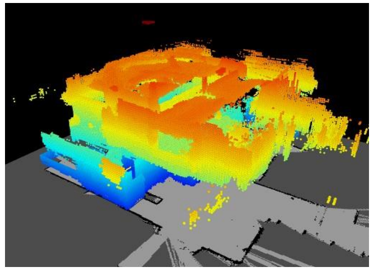
电力巡检

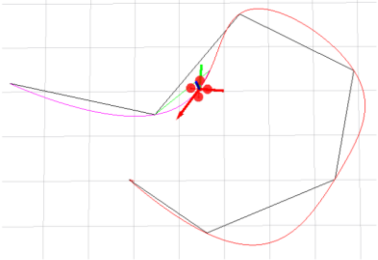
行业应用

无人配送

执法

无人机自主导航架构





传感器数据
- IMU数据
- 点云数据

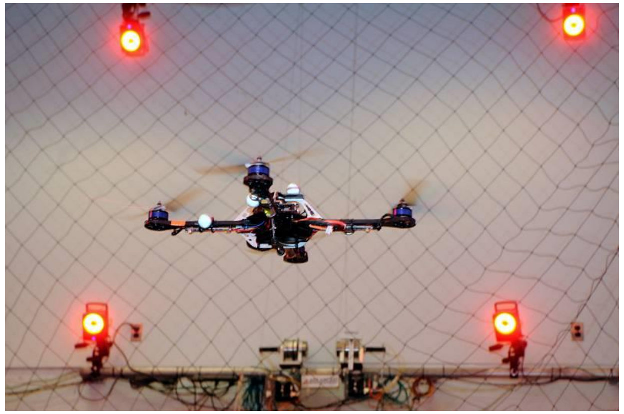
定位与建图（SLAM）
- 定位
  - 视觉：VIO
  - 雷达：LIOSAM
- 建图
  - 滚动栅格地图
  - ESDF

运动规划
- 前端：路径搜索/采样飞行走廊生成
- 后端：轨迹优化

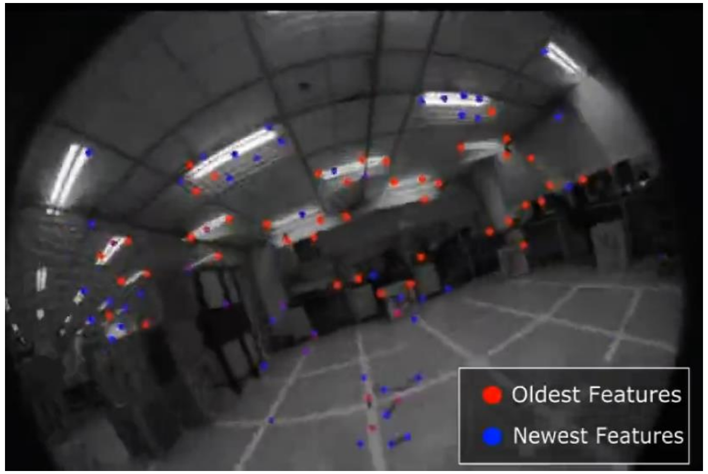
控制器
- 位置、速度闭环
- 姿态控制

定位

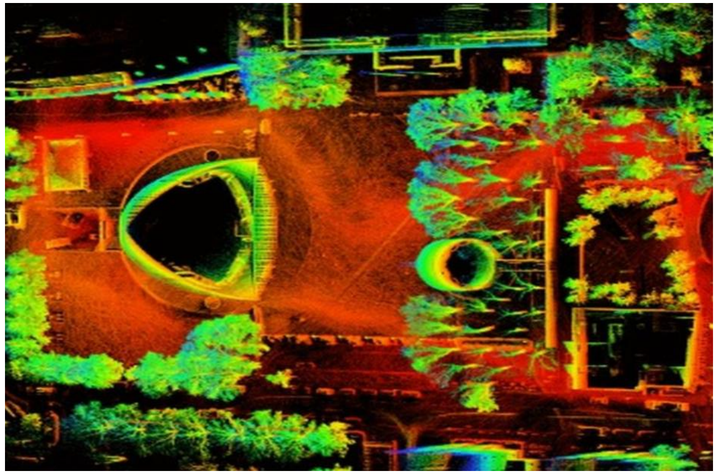
GPS



动作捕捉仪

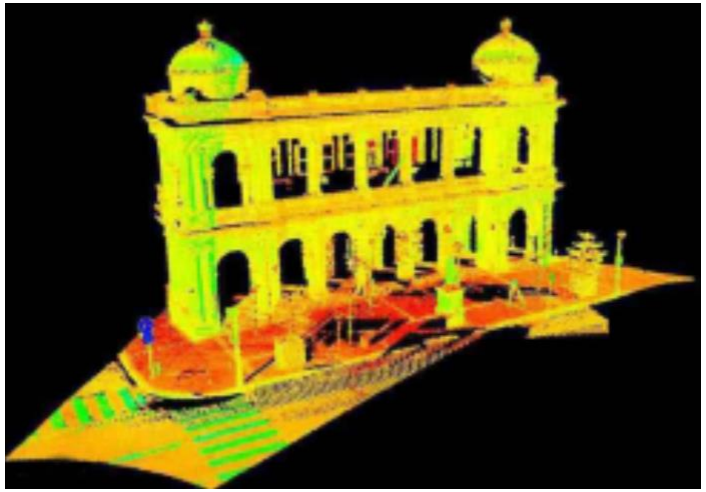


视觉惯性里程计（VIO）

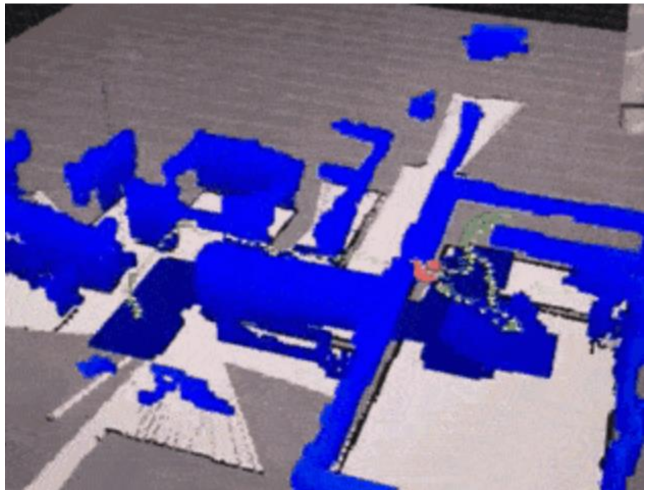


激光惯性里程计

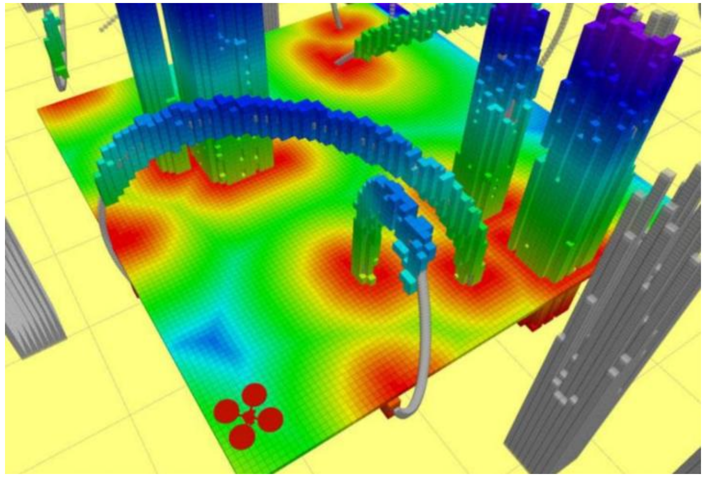
建图



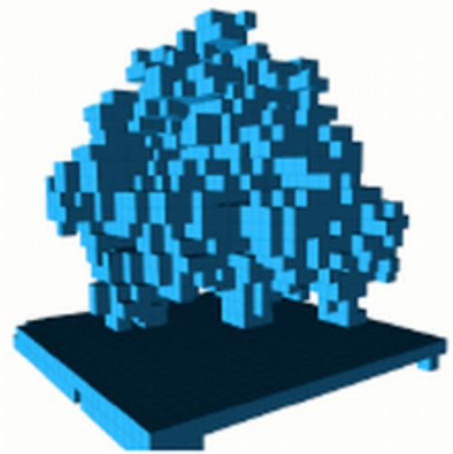
点云地图



栅格地图

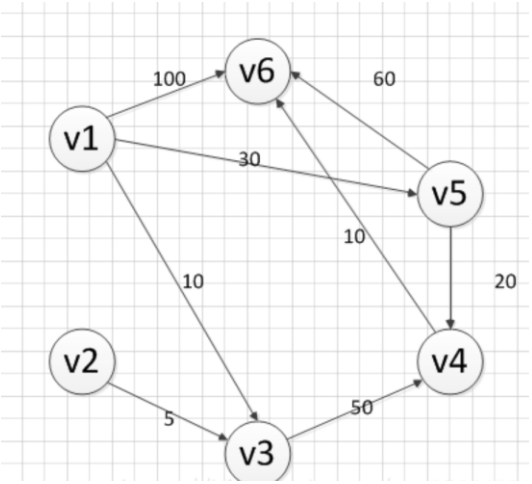


欧氏距离场(ESDF)

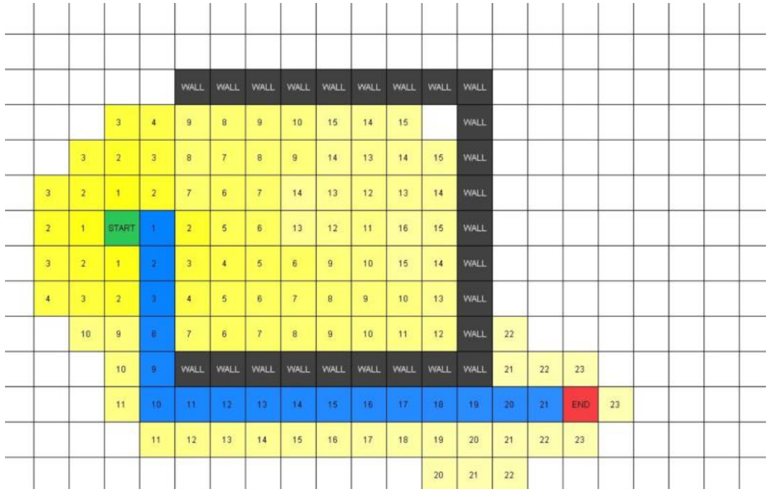


八叉树地图

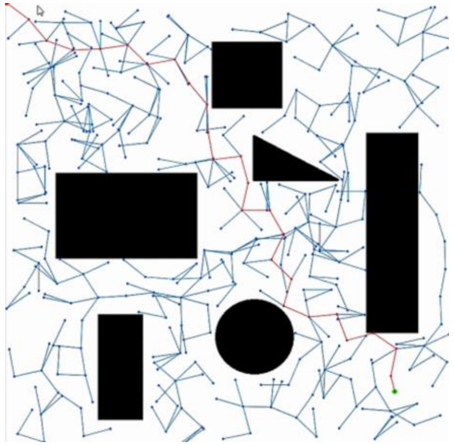
路径搜索



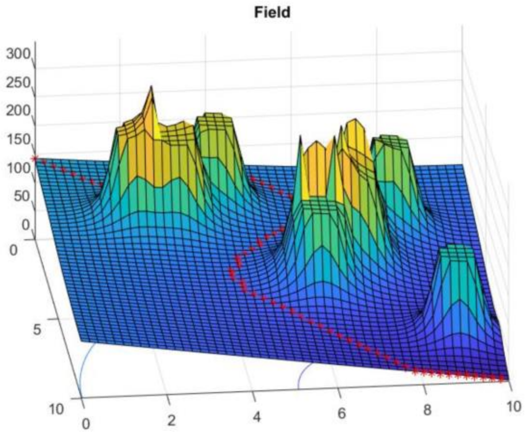
Dijkstra



Astar

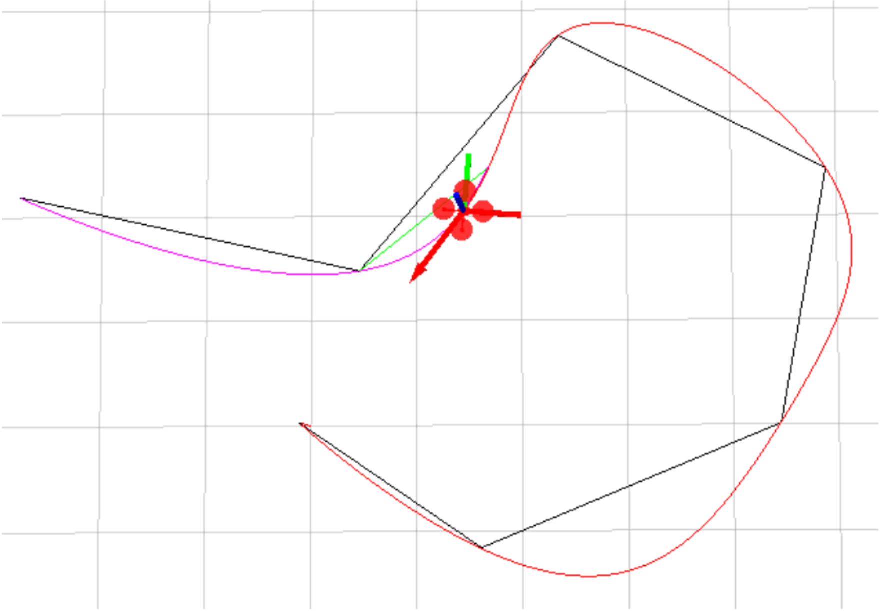


RRT



人工势场法

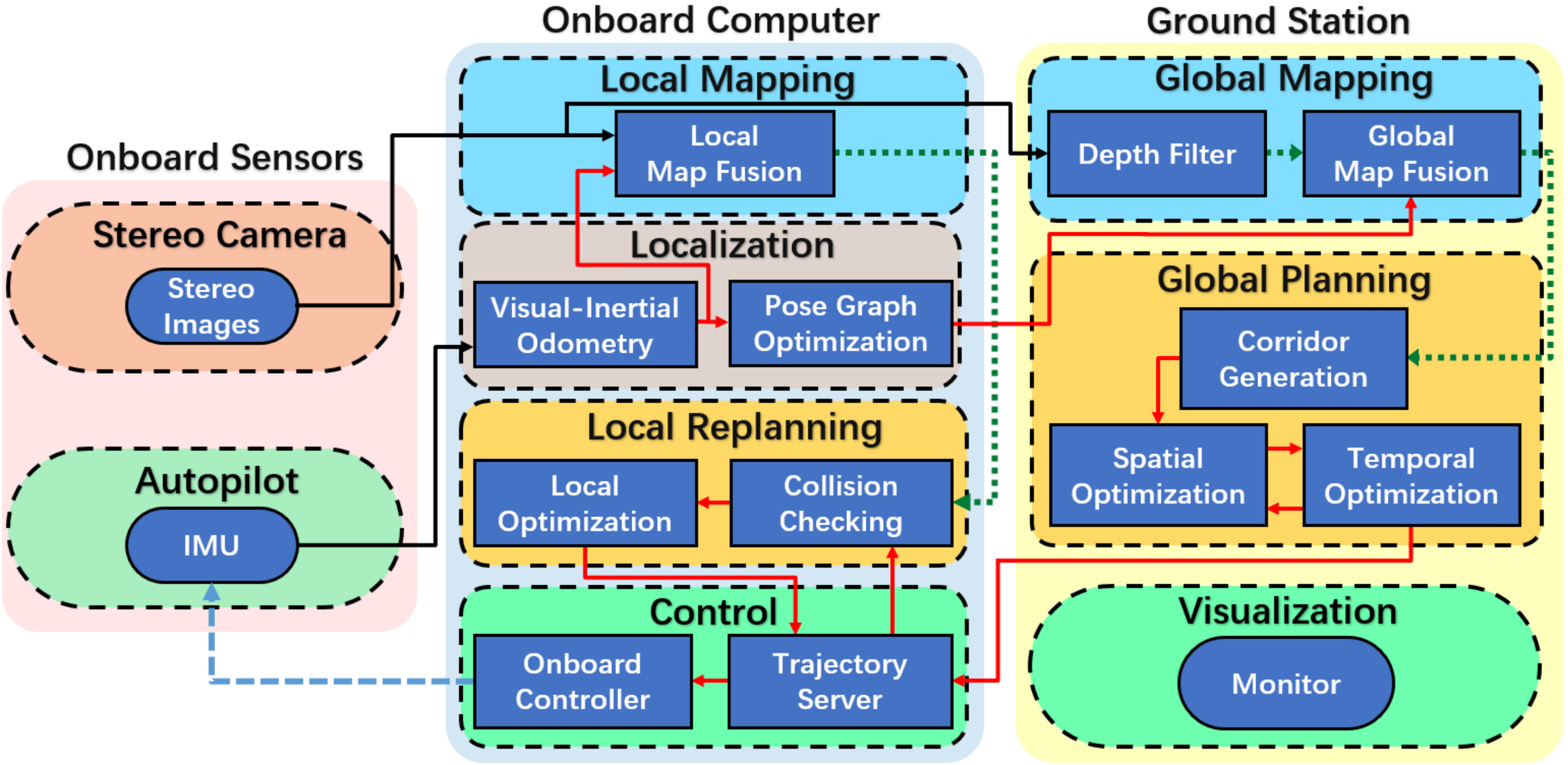
轨迹优化



Minimum-Snap

SE(3)轨迹优化

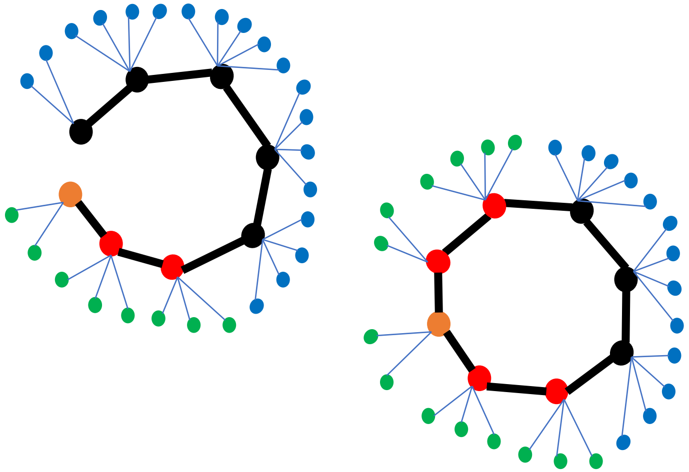
Teach-Repeat-Replan System: Overview

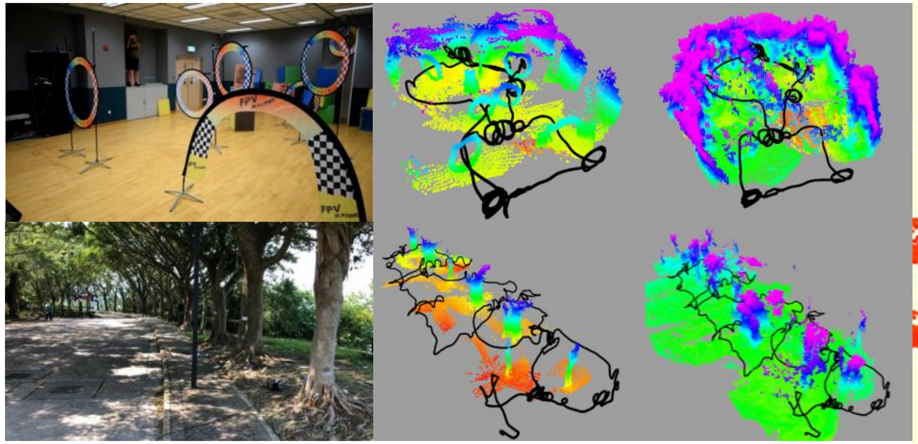


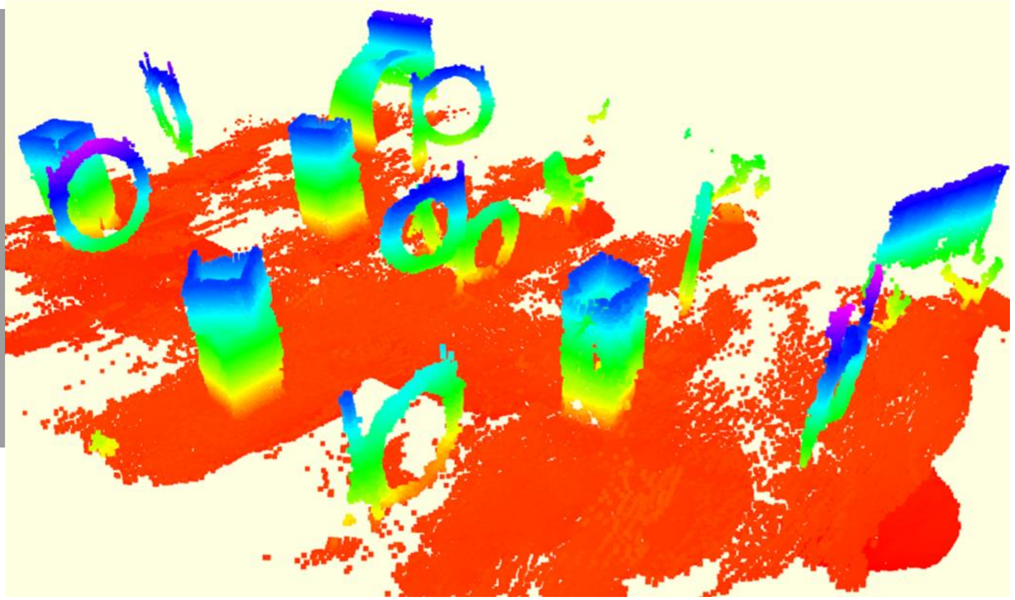
Teach-Repeat-Replan: A Complete and Robust System for Aggressive Flight in Complex Environments, Fei Gao, Shaojie Shen et al., IEEE Transactions on Robotics(**TRO**) 2020.
King-Sun Fu Memorial Best Paper Award Honorable Mention.

GitHub Repo

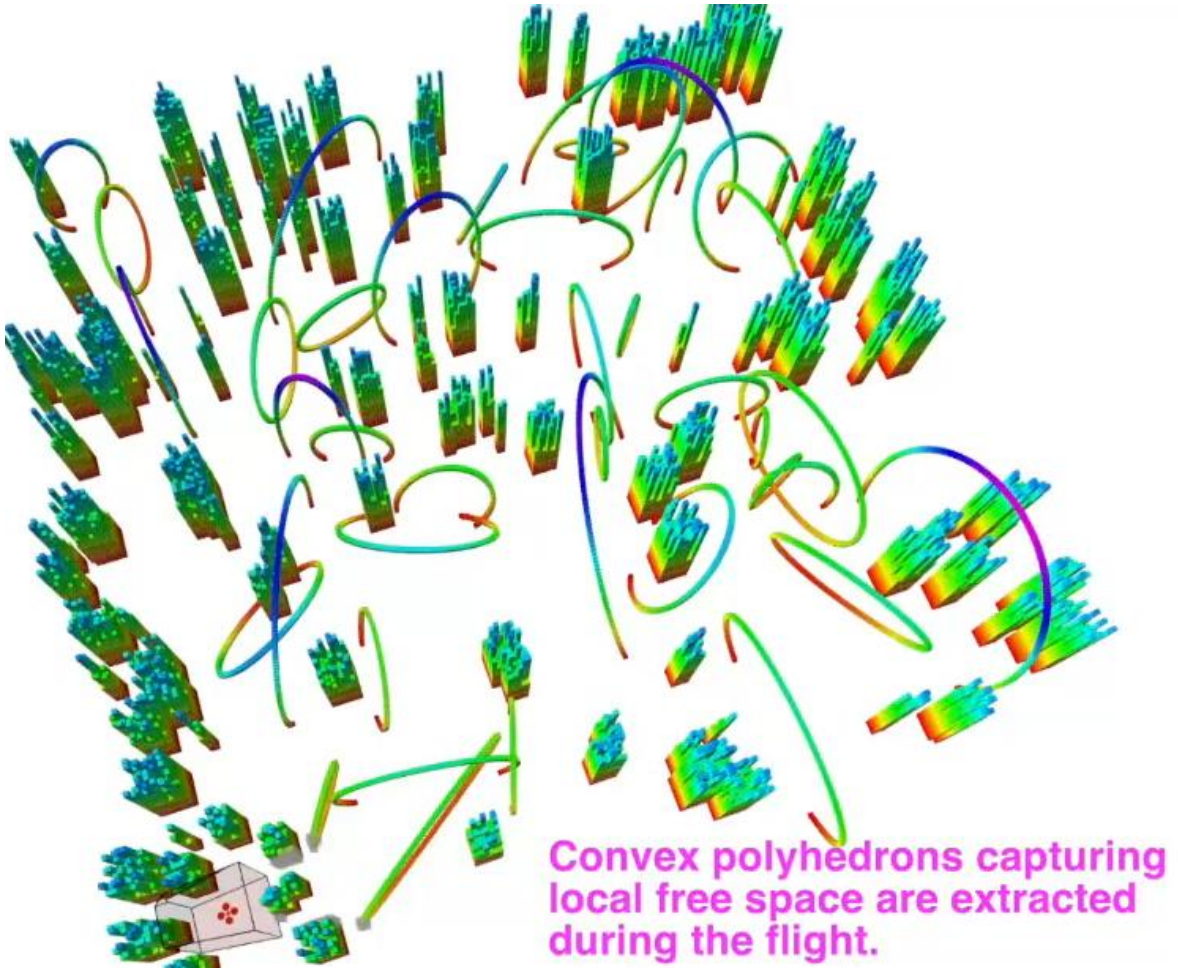
Teach-Repeat-Replan System: Global Mapping

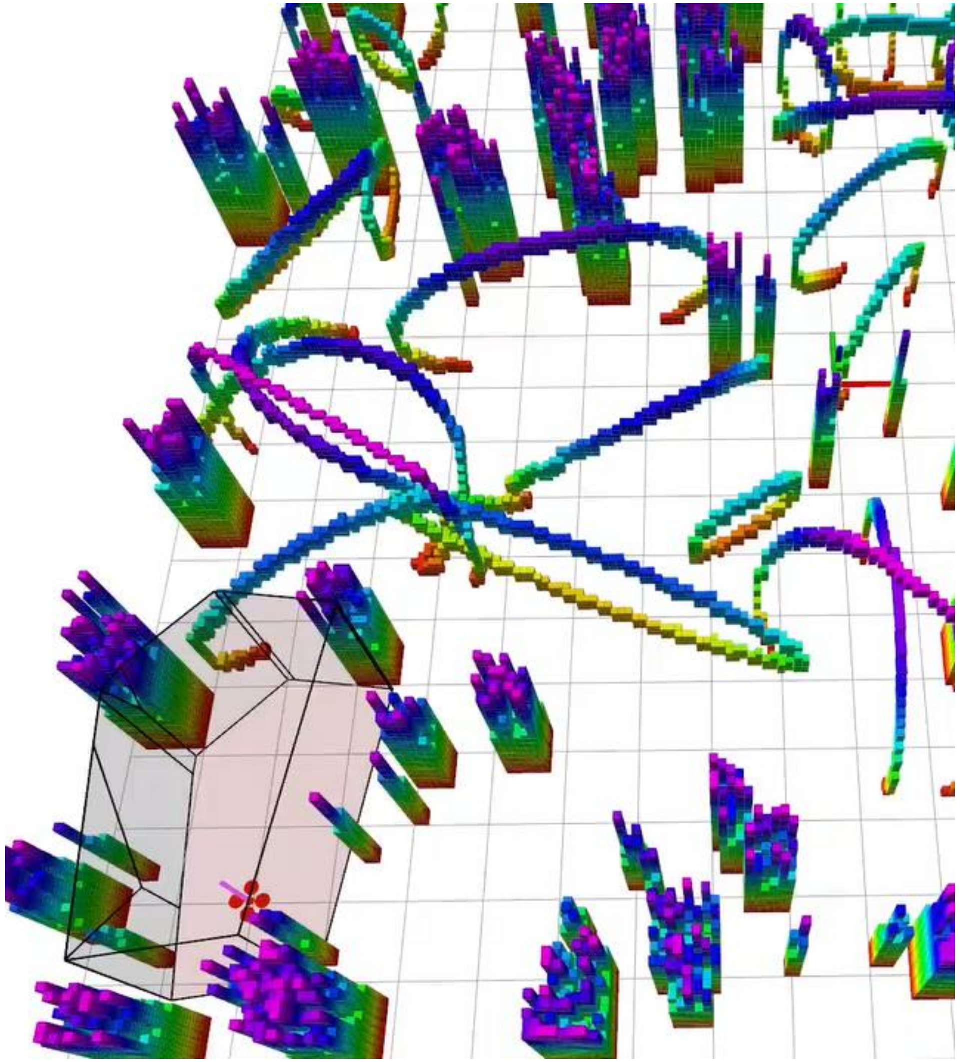




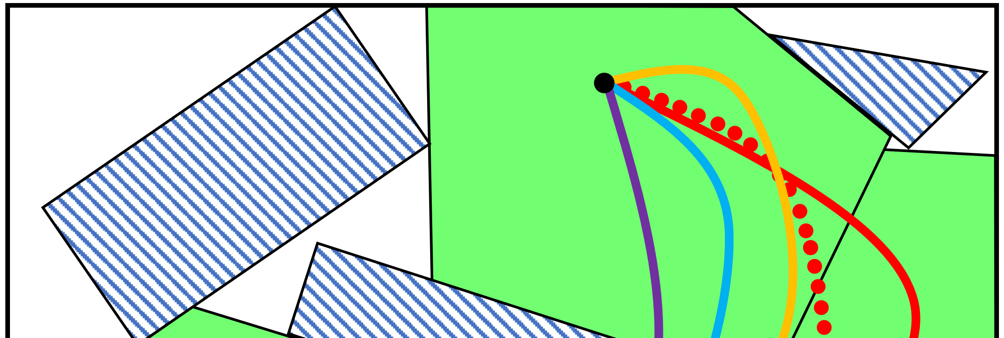


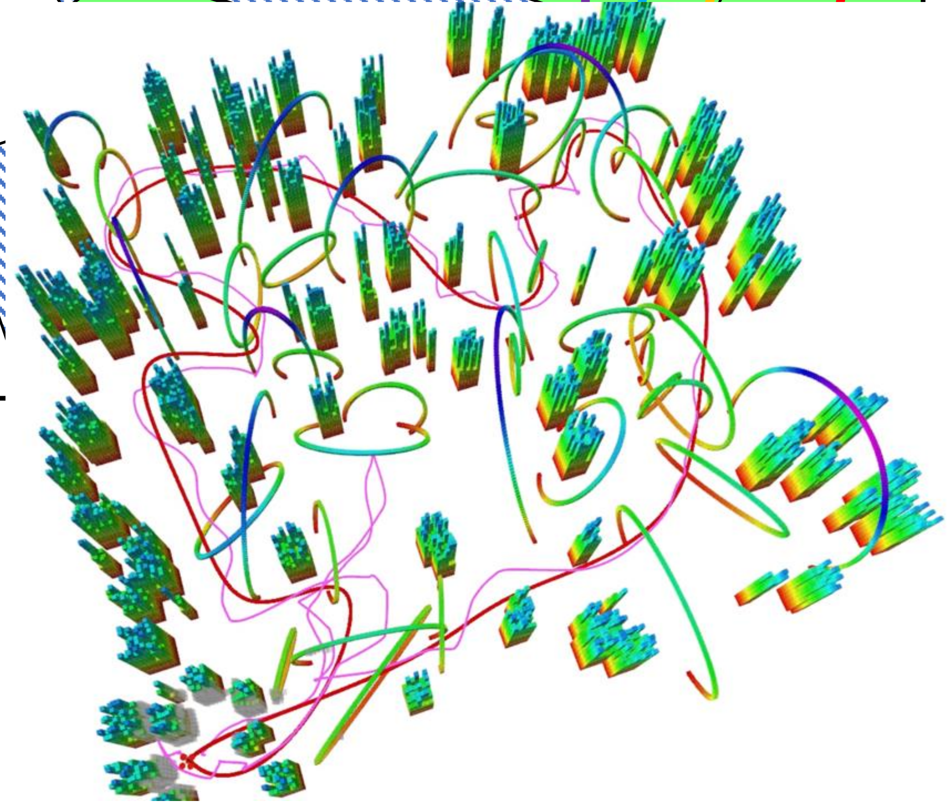
Teach-Repeat-Replan System: Flight Space Generation





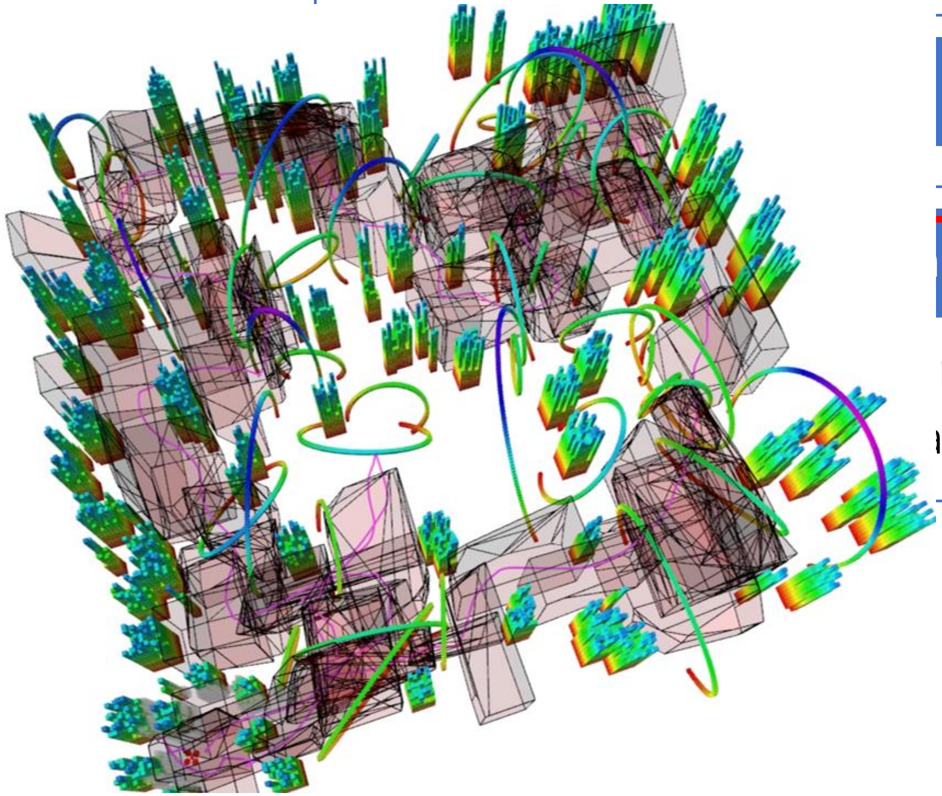
Teach-Repeat-Replan System: Global Trajectory Optimization





User gives a teach path

Find a Topo-equivalent Corridor



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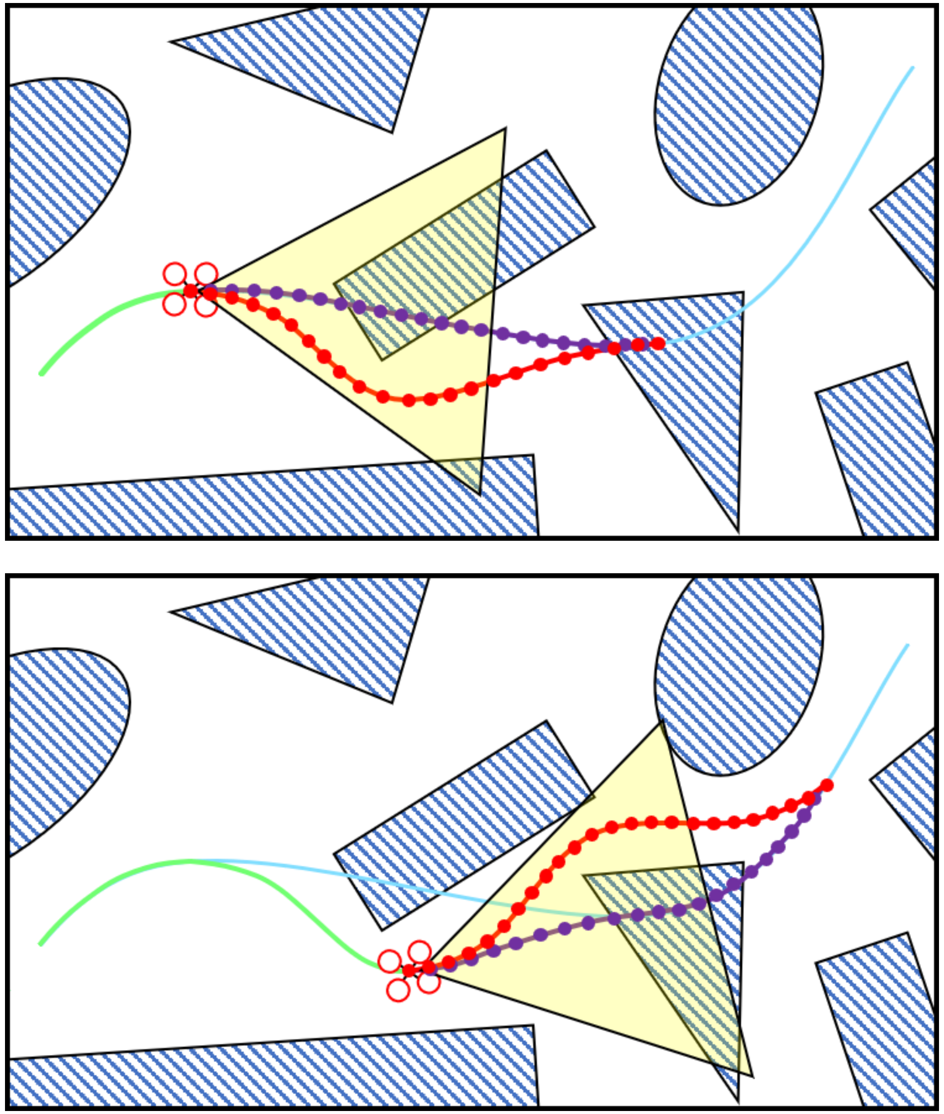
Teach-Repeat-Replan System: Local Planning

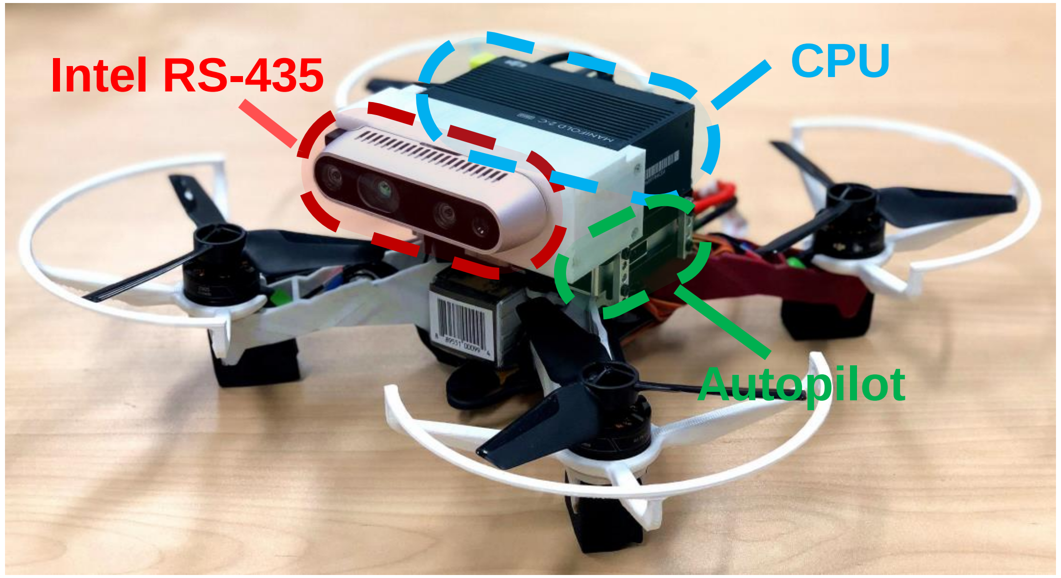
- Compare our system FIESTA with Voxblox
- Intel i7-8700k at **3.7GHz** and only 1 thread is used

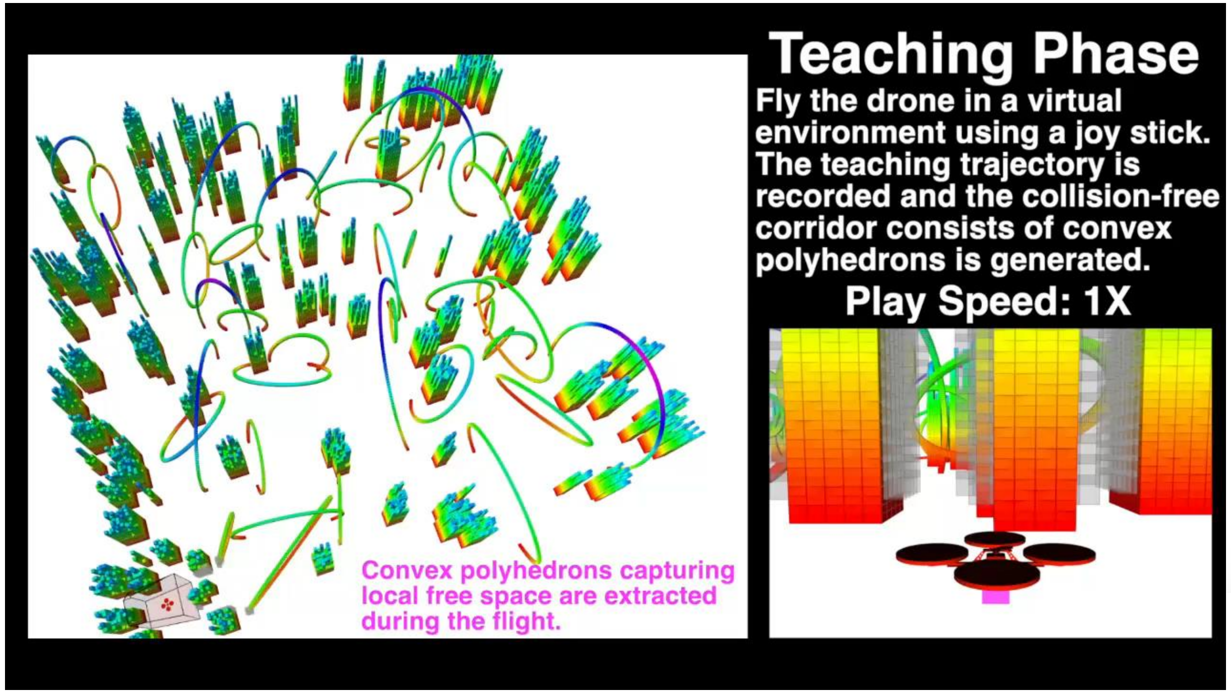
Resolution: 0.1m　　　Resolution: 0.2m

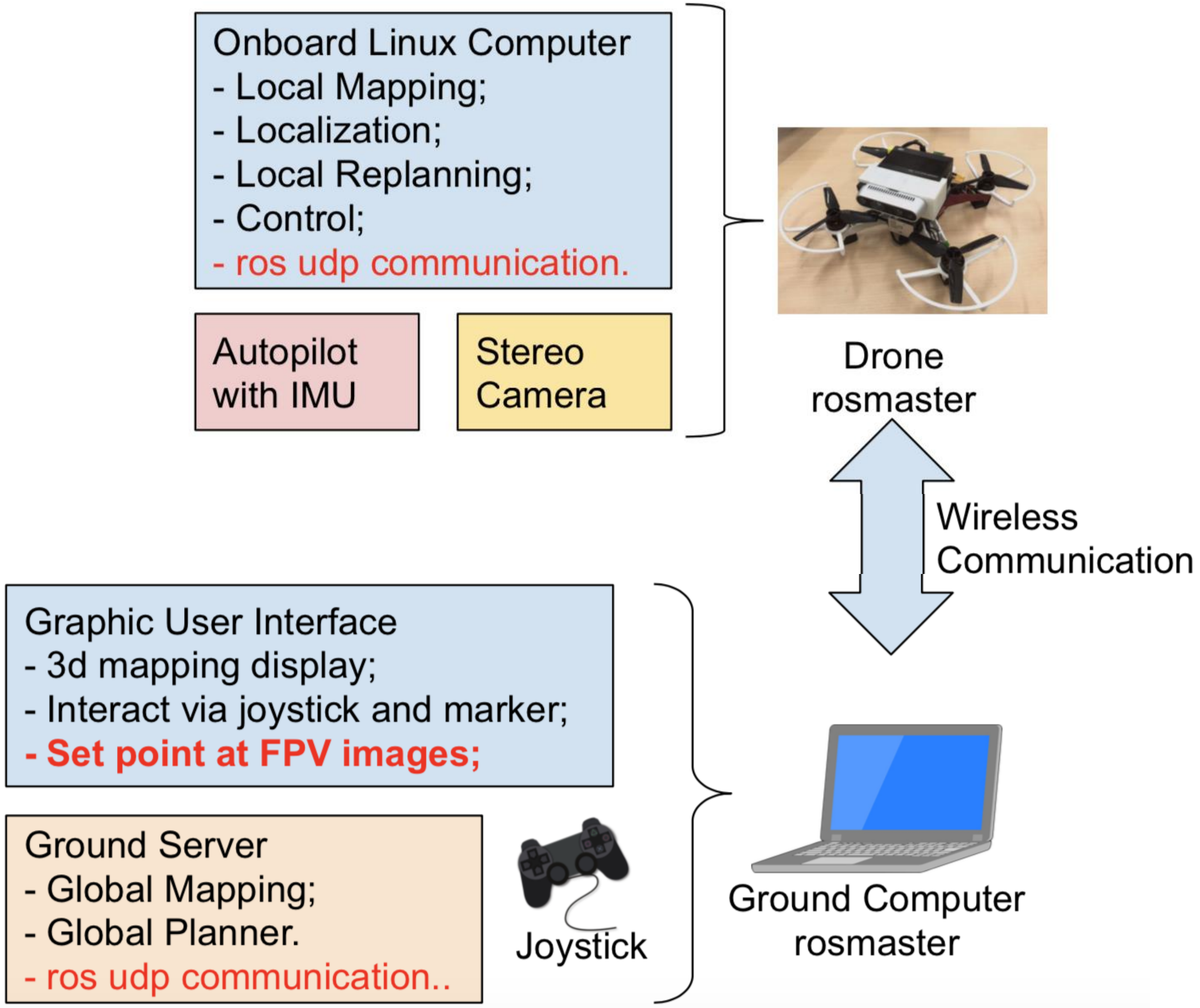
The cubes constitutes all the obstacles, while the plane is a slice of ESDF at z=0.7

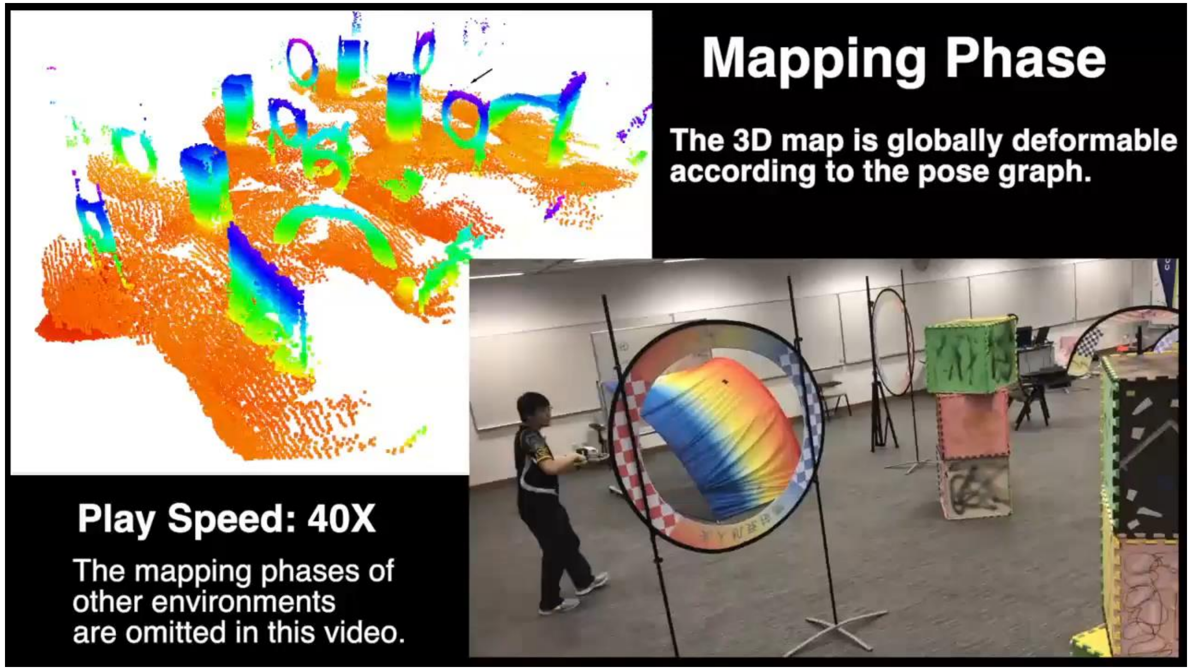


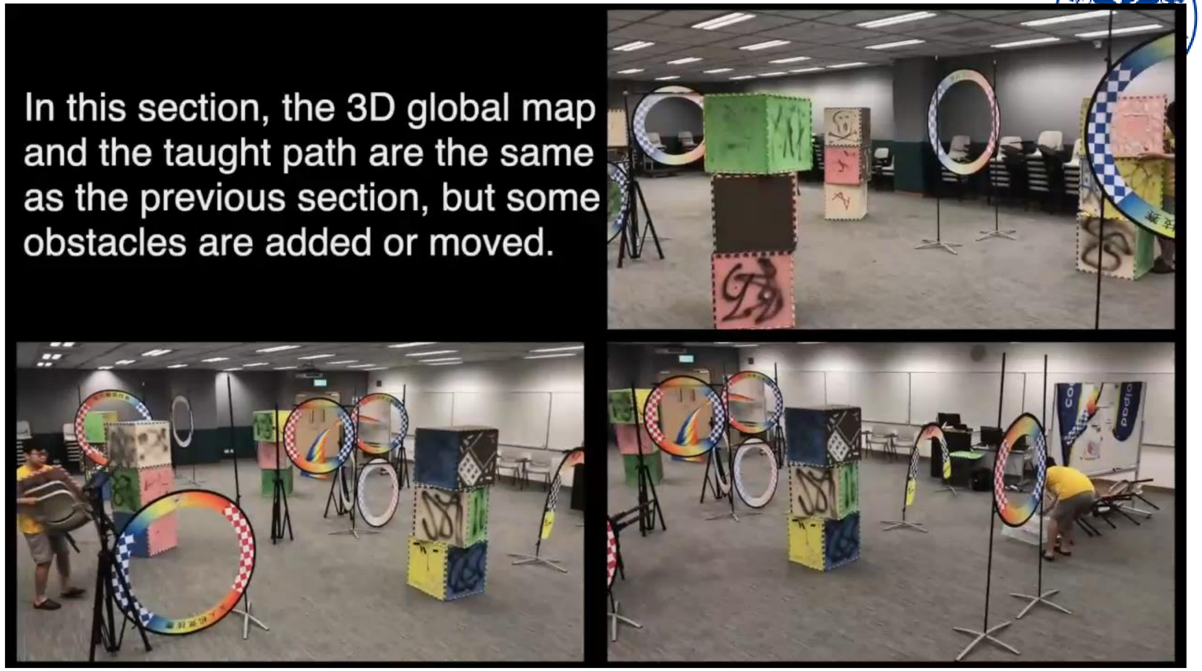
Teach-Repeat-Replan System: Integration











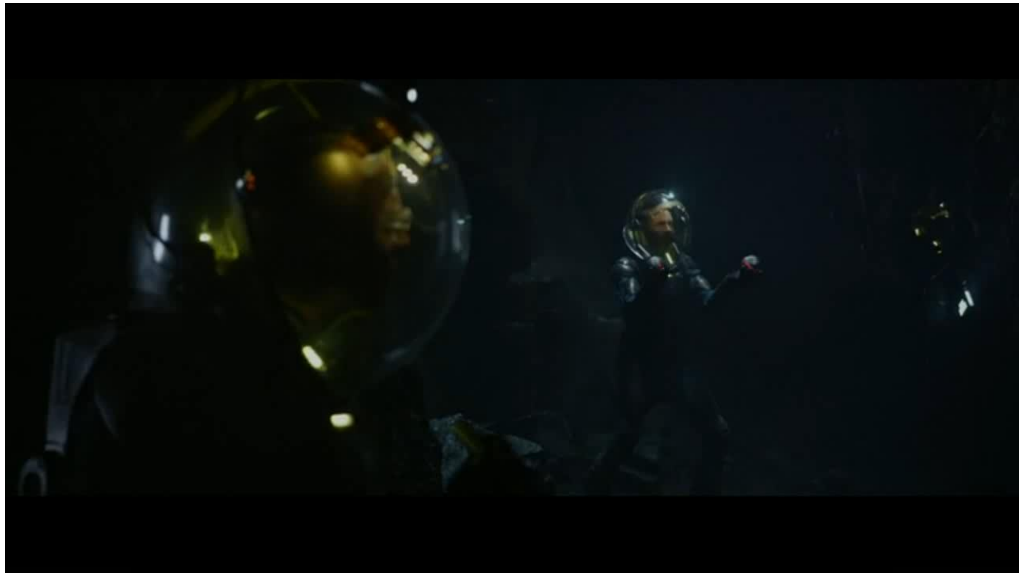
Comparison with A Skilled Pilot

Challenges

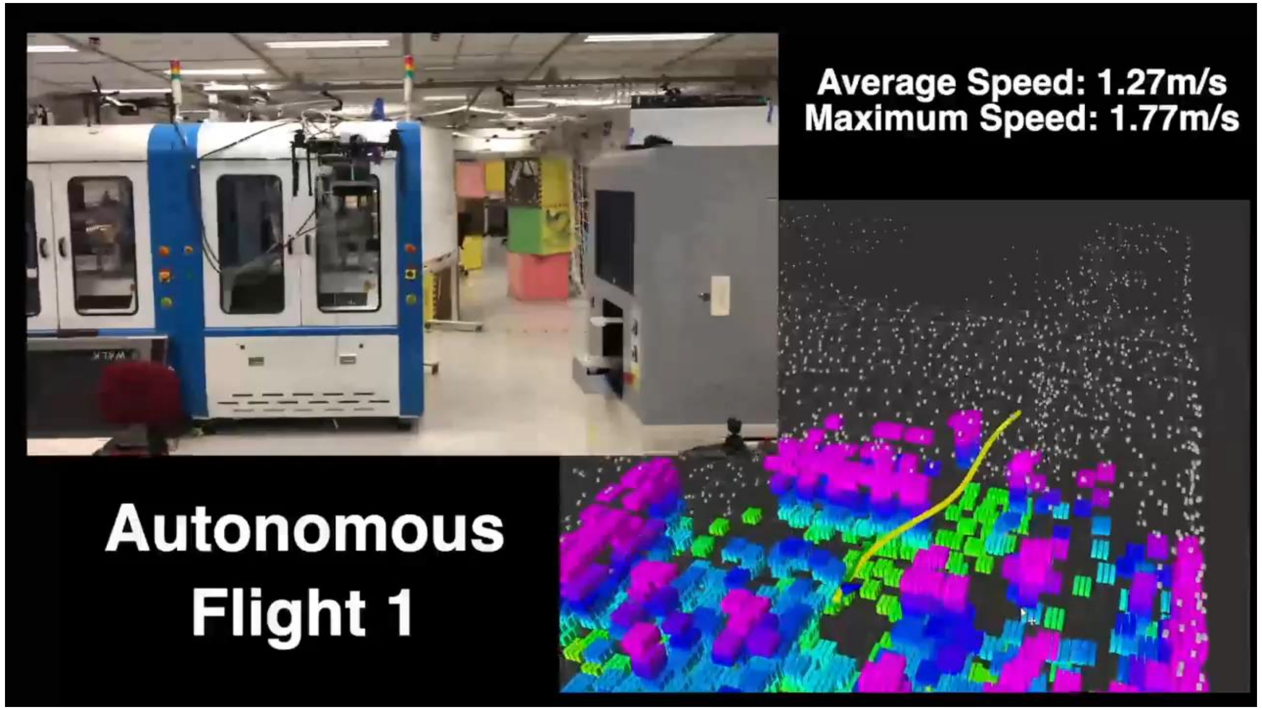
- **Lightweight**
  - Cut-off redundant parts while retain good performance.
- **Robust**
  - Resilient to disturbance with safety guarantee.
- **Agile**
  - Real-time feasibility constrained SE(3) planning.

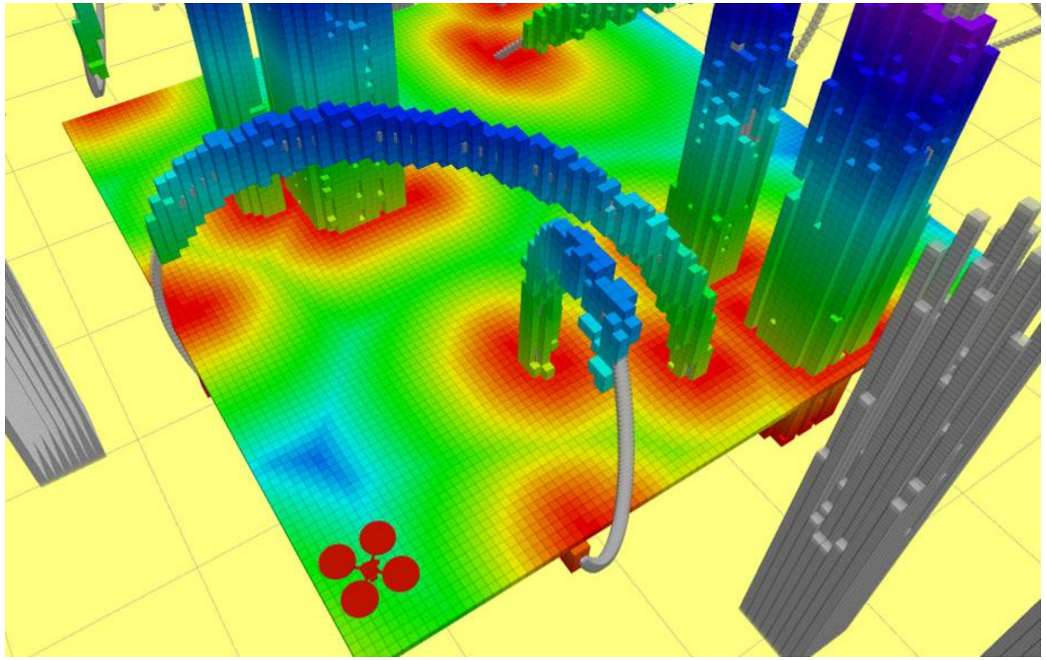
Our Ultimate Goal:

Tiny, Intelligent, Cooperative, Decentralized Swarm



Lightweight: ESDF-free Gradient-based Collision Avoidance





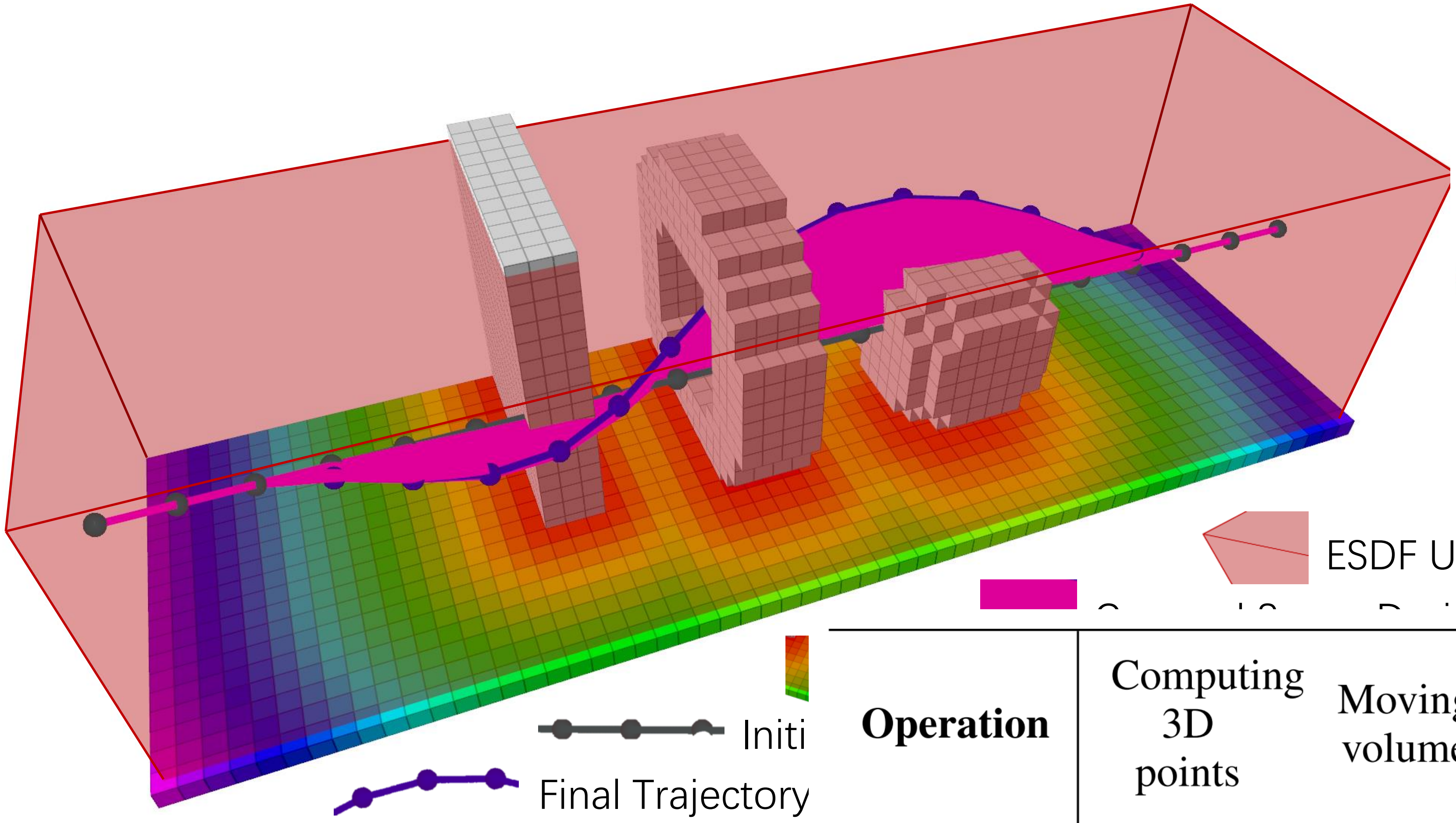
- Gradient methods naturally suit local (re)planning.
- Advancements are witnessed in past years.
- Can be robust and efficient.

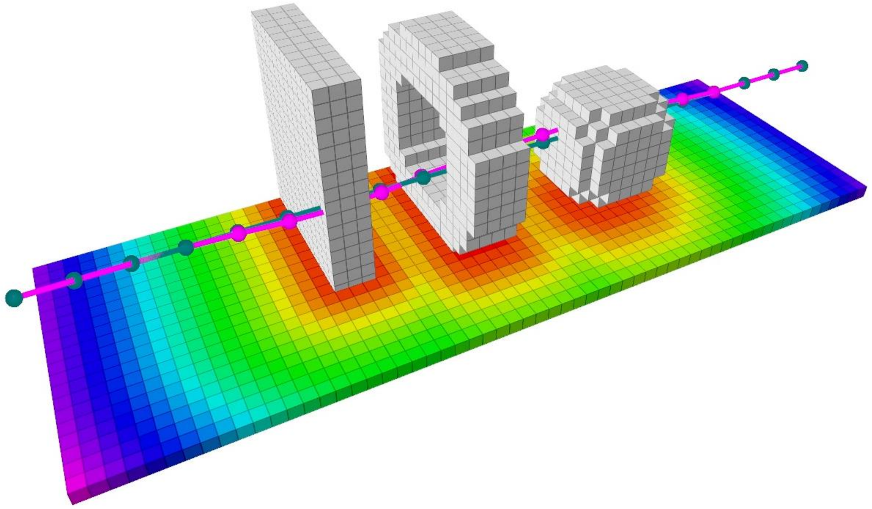
ESDF map
For evaluation of the gradient magnitude and direction

Thinking: Is ESDF truly necessary?

Robust and Efficient Quadrotor Trajectory Generation for Fast Autonomous Flight, Boyu Zhou, Fei Gao, et al., IEEE Robotics and Automation Letter 2019.
FIESTA: A Fast Incremental Euclidean Distance Fields for Online Quadrotor Motion Planning, Luxin Han, Fei Gao, et al., IEEE/RSJ IROS 2019.

Lightweight Local Collision Avoidance



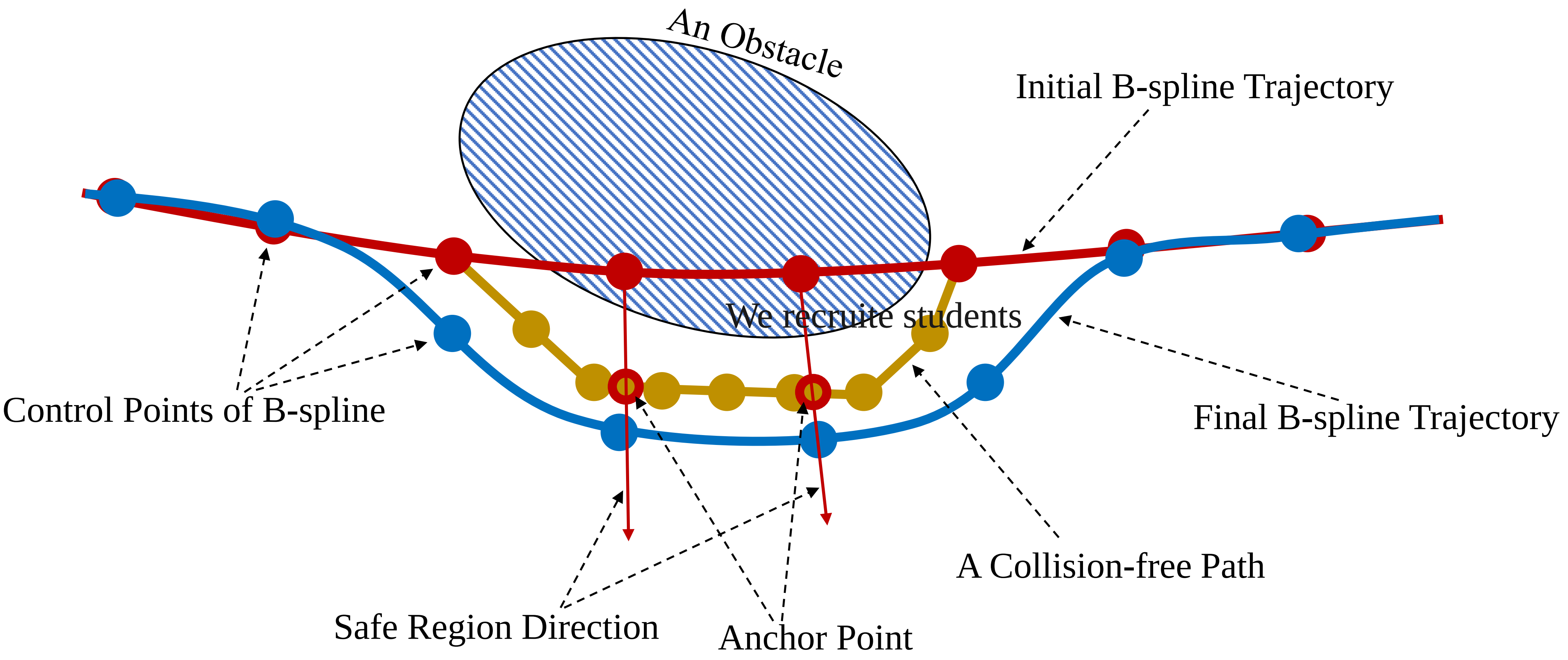


Operation	Computing 3D points	Moving volume	Inserting mea-sure-ments	SDF compu-tation	Trajectory opti-mization
Time [ms]	0.265	0.025	0.518	9.913	3.424

Key Insight: The trajectory during optimizi
much more limited space than the ESDF does.

Real-Time Trajectory Replanning for MAVs using Uniform B-splines and a 3D Circular Buffer, V. Usenko et al, IROS 2017

ESDF-free Gradient-based Collision Avoidance



The trajectory is parameterized by a uniform B-spline with control points Q

Solve: $\min_{Q} J = \lambda_s J_s + \lambda_c J_c + \lambda_d J_d$

J_s is smoothness penalty
$J_c$is for collision
J_c indicates dynamic feasibility

ESDF-free Gradient-based Collision Avoidance

Step 1: Optimize

w_s, w_c, w_d: weights

$$\min_{\mathbf{Q}}\{w_s \cdot J_s(\mathbf{Q}) + w_c \cdot J_c(\mathbf{Q}) + w_d \cdot J_d(\mathbf{Q})\}$$

Q: trajectory

$J_c(\mathbf{Q})$: collision penalty

$J_s(\mathbf{Q})$: smoothness penalty

$J_c(\mathbf{Q})$: dynamical feasibility penalty

Collision Still Detected!

Step 2: Check collision again

Safe

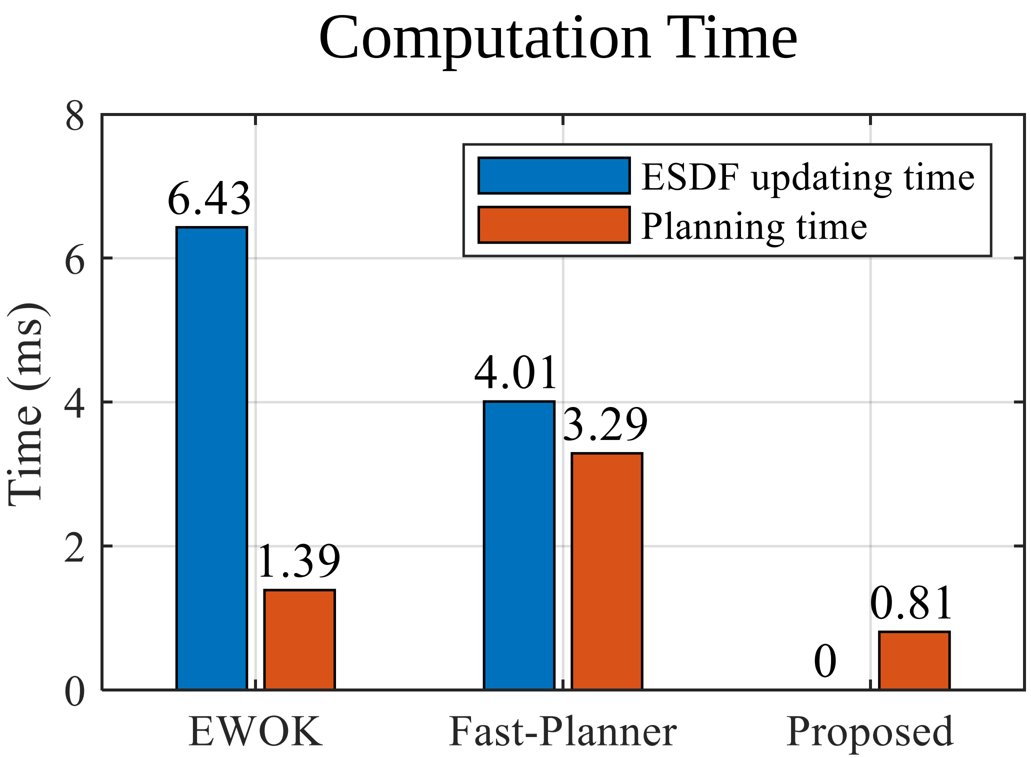
Success!

ESDF-free Gradient-based Collision Avoidance

1. Trajectory moves away from obstacles without ESDF.
2. Computation time is reduced by only operating the necessary obstacles.

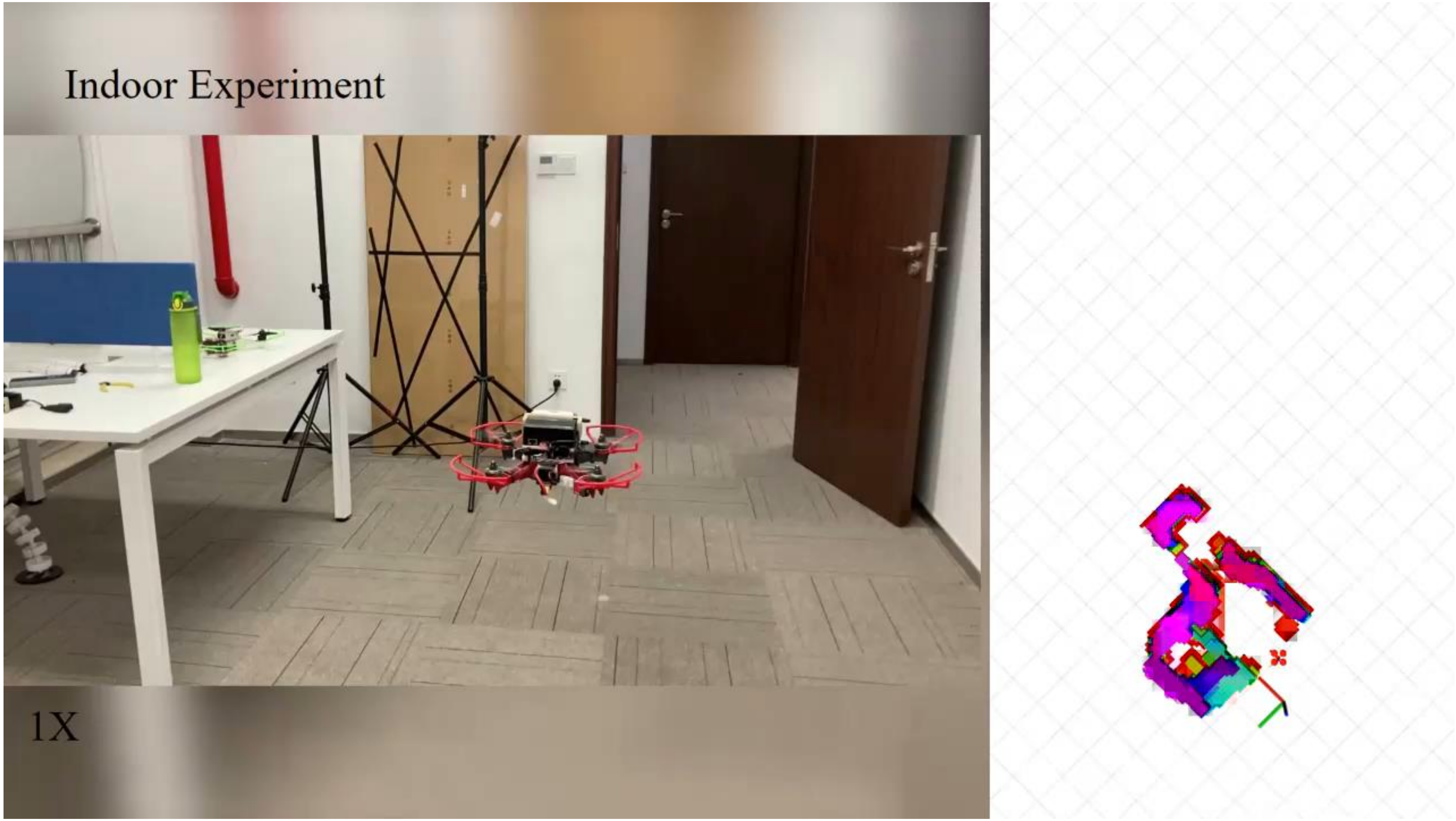
ESDF-free Gradient-based Collision Avoidance

Significantly reduces the computation time while achieving impressive flight performance.



EGO-Planner: An ESDF-free Gradient-based Local Planner for Quadrotors, Xin Zhou, Fei Gao, et al., IEEE Robotics and Automation Letter 2021.

ESDF-free Gradient-based Collision Avoidance



EGO-Planner: An ESDF-free Gradient-based Local Planner for Quadrotors, Xin Zhou, Fei Gao, et al., IEEE Robotics and Automation Letter 2021.

Robust: Disturbance-aware Navigation

- Objective & Constraints

State: $\mathbf{x}^{(k)} = \left[x, v_x, a_x, y, v_y, a_y, z, v_z, a_z, \theta, v_\theta, a_\theta\right]^T$

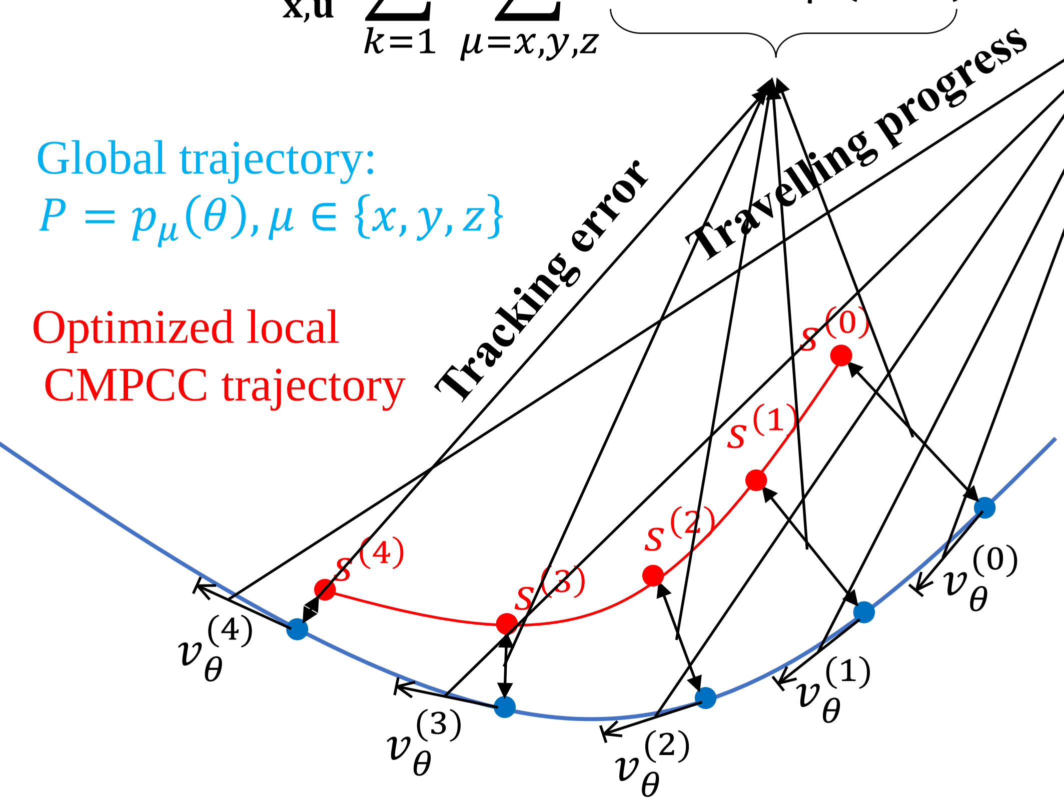
Input: $\mathbf{u}^{(k)} = \left[j_x, j_y, j_z, j_\theta\right]^T$

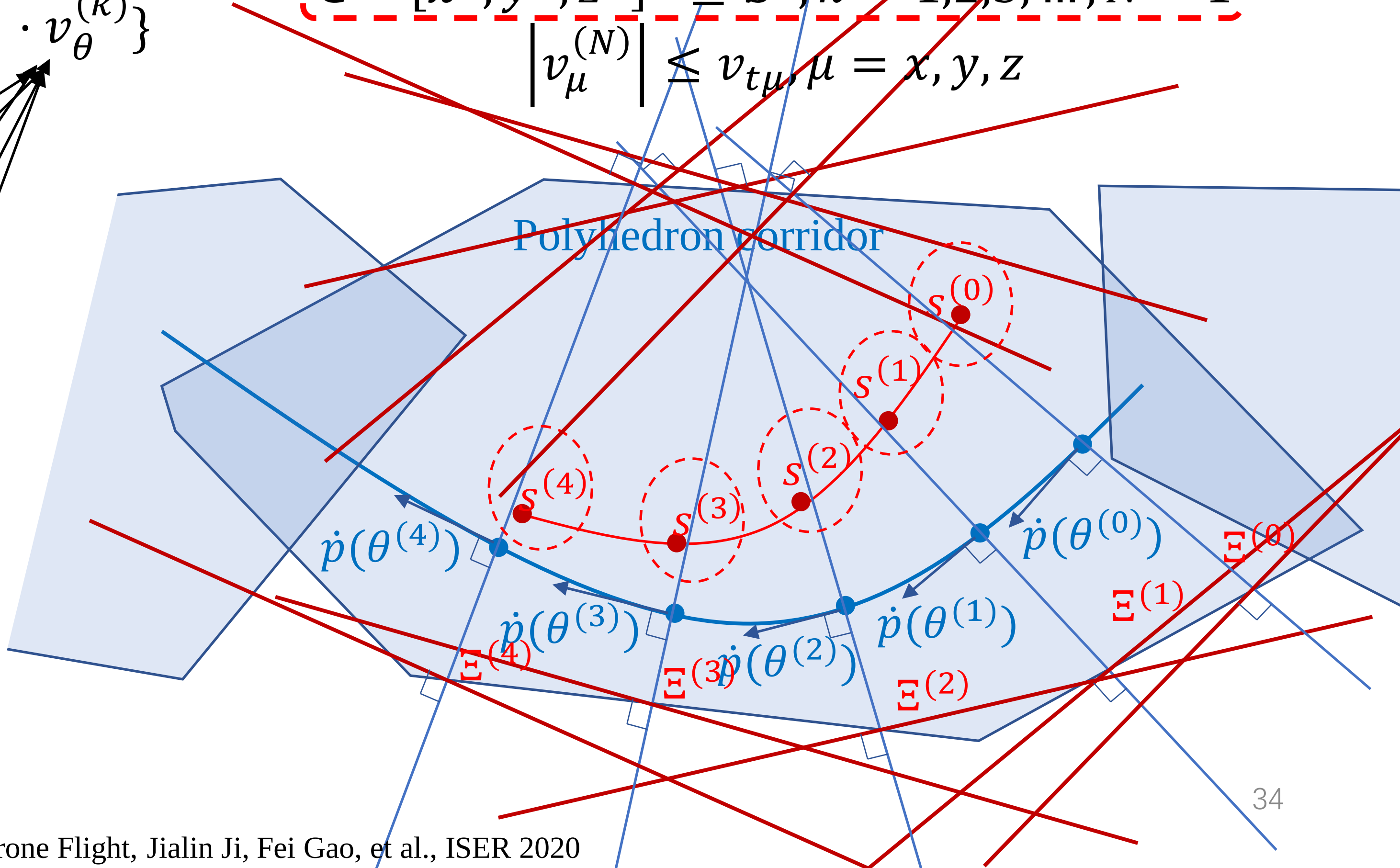
$$J = \min_{\mathbf{x},\mathbf{u}} \sum_{k=1}^{N} \{ \sum_{\mu=x,y,z} \underbrace{\left(\mu^{(k)} - p_\mu\left(\theta^{(k)}\right)\right)^2} - q \cdot v_\theta^{(k)} \}$$

$$\begin{aligned} \text{s.t. } \mathbf{x}^{(k+1)} &= \mathbf{A_d}\mathbf{x}^{(k)} + \mathbf{B_d}\mathbf{u}^{(k)}, k = 1,2,3, \dots, N-1 \\ \mathbf{x}_l \le \mathbf{x}^k &\le \mathbf{x}_u, k = 1,2,3, \dots, N-1 \\ \mathbf{u}_l \le \mathbf{u}^k &\le \mathbf{u}_u, k = 1,2,3, \dots, N-1 \\ \boxed{\mathbf{C}^k \cdot [\boldsymbol{x}^k, \boldsymbol{y}^k, \boldsymbol{z}^k]^T \le \mathbf{b}^k, k = 1,2,3, \dots, N-1} \\ \left|v_\mu^{(N)}\right| &\le v_{t\mu}, \mu = x, y, z \end{aligned}$$

Global trajectory:
$P = p_\mu(\theta), \mu \in \{x, y, z\}$

Optimized local
CMPCC trajectory

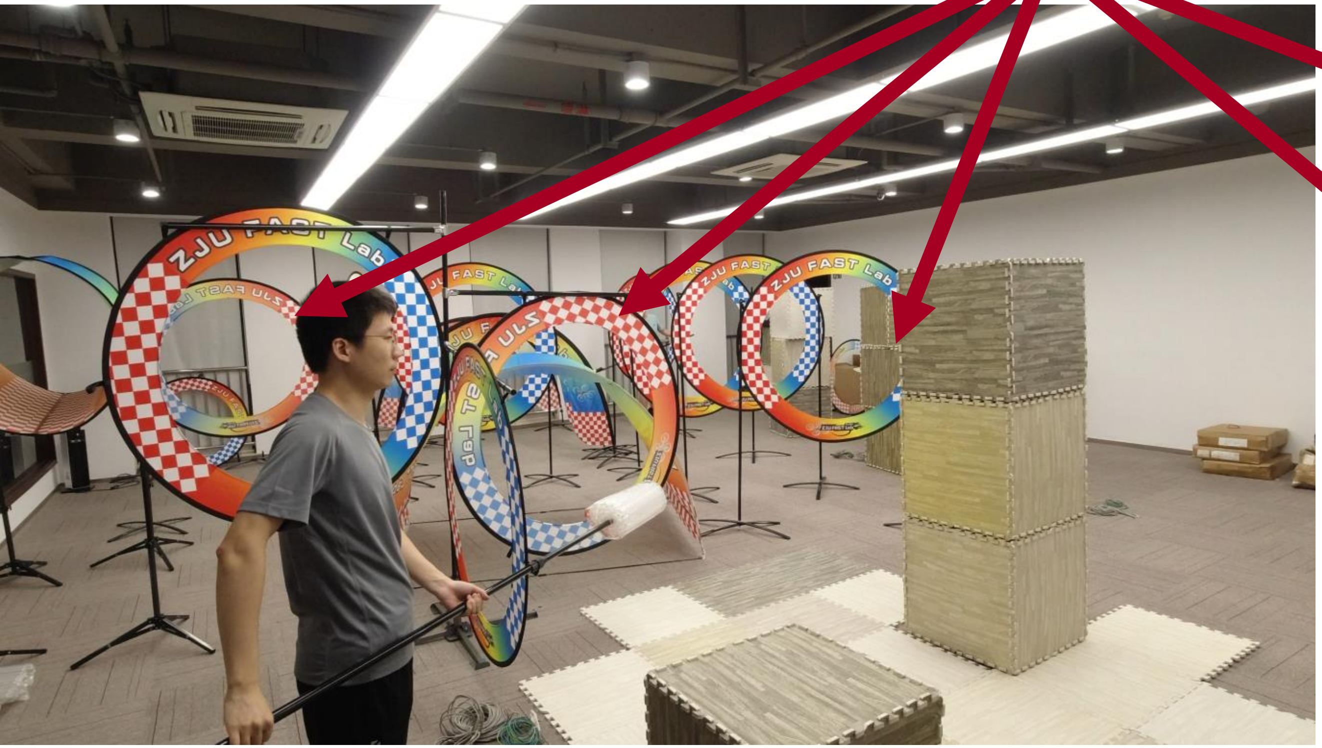


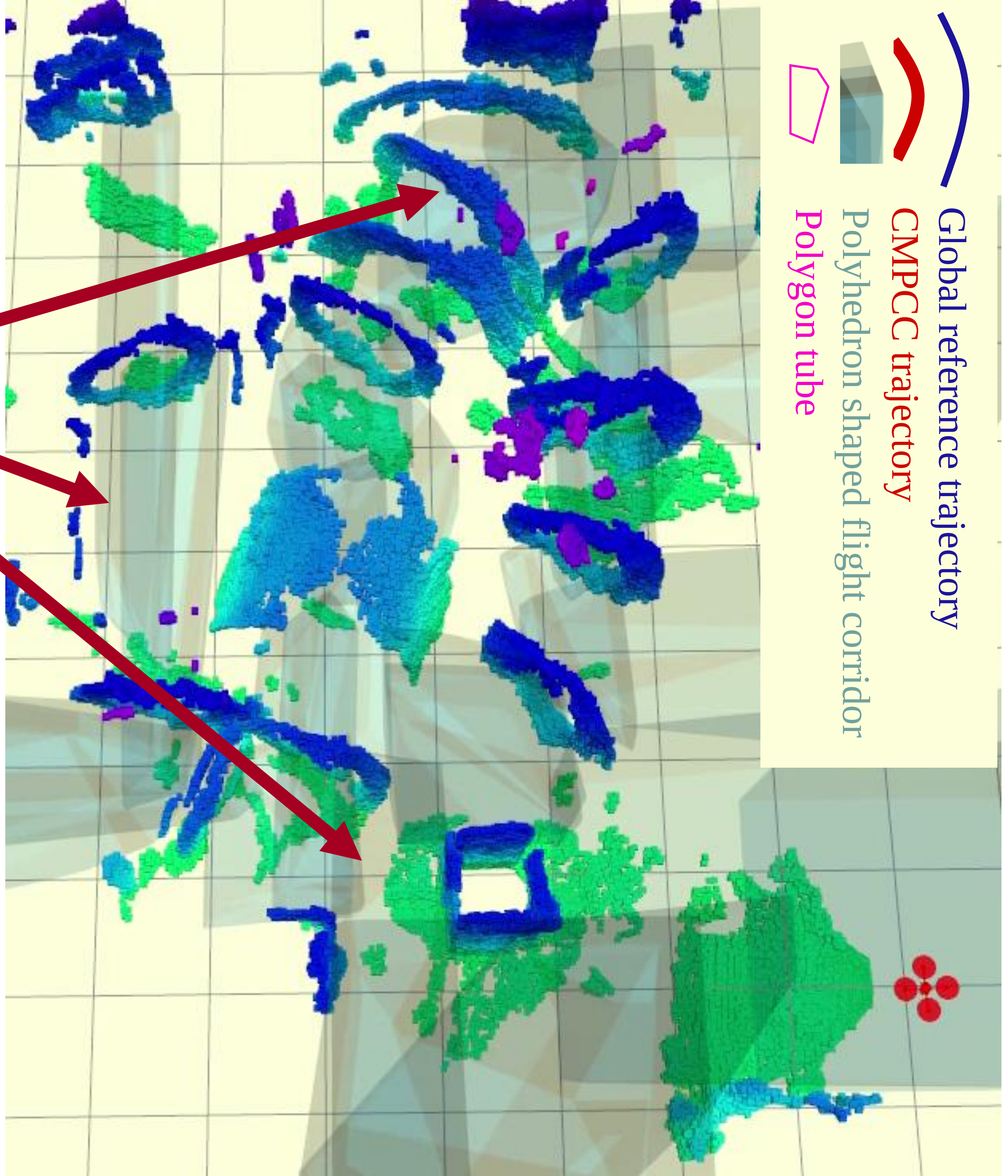


CMPCC: Corridor based Model Predictive Contouring Control for Aggressive Drone Flight, Jialin Ji, Fei Gao, et al., ISER 2020

Disturbance-aware Navigation

Disturbance occur!





All videos play without speed up. This is the same experiment in previous video with fixed view.

Disturbance-aware Navigation

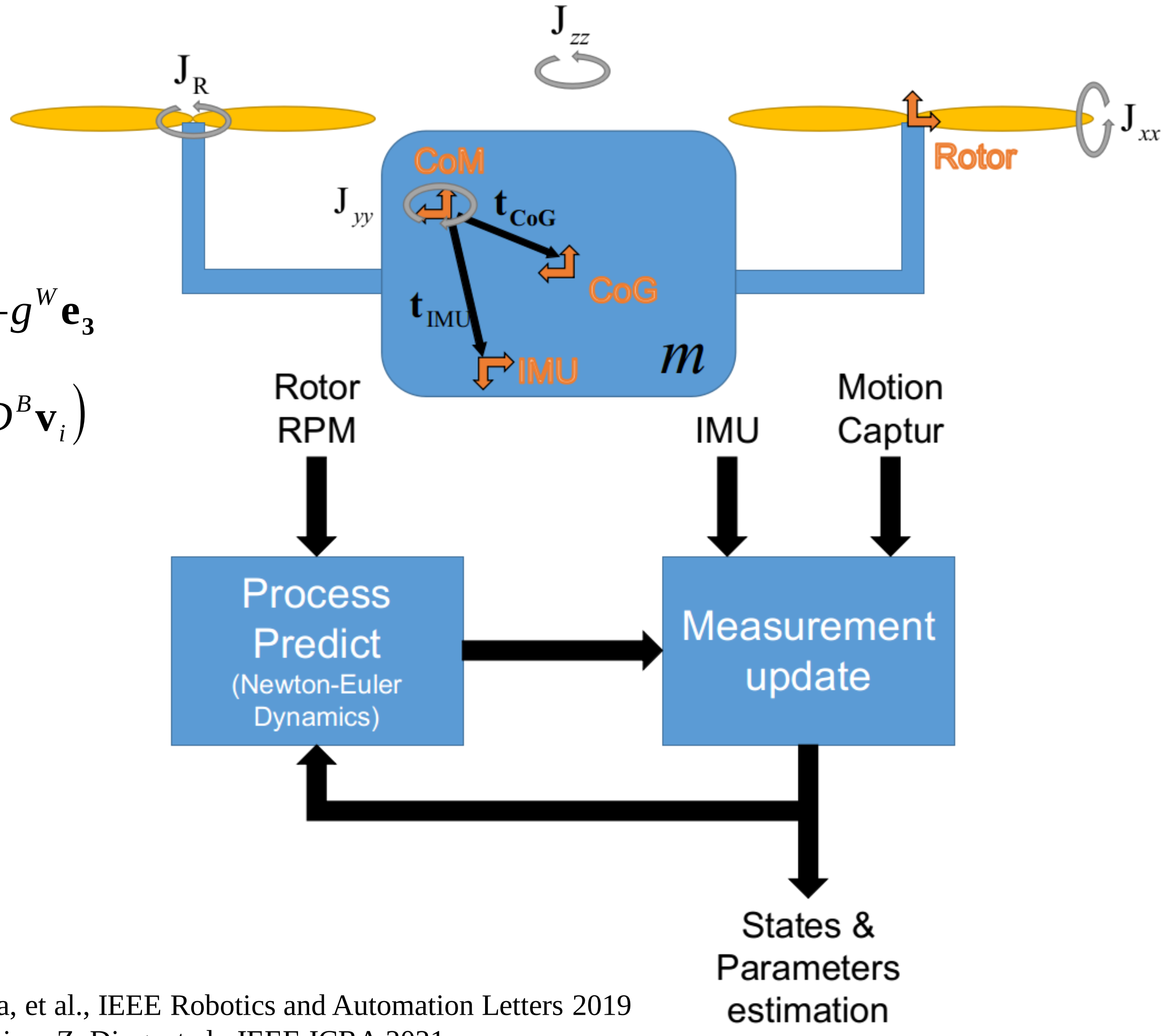
External forces caused by air turbulence, loaded objects, or other uncertain disturbance challenge the stability of planning.

External Forces Resilient Safe Motion Planning for Quadrotor, Y. Wu, et al., IEEE Robotics and Automation Letters 2021

Disturbance-aware Navigation

➢ Nonlinear System Identification Using Newton-Euler Dynamics

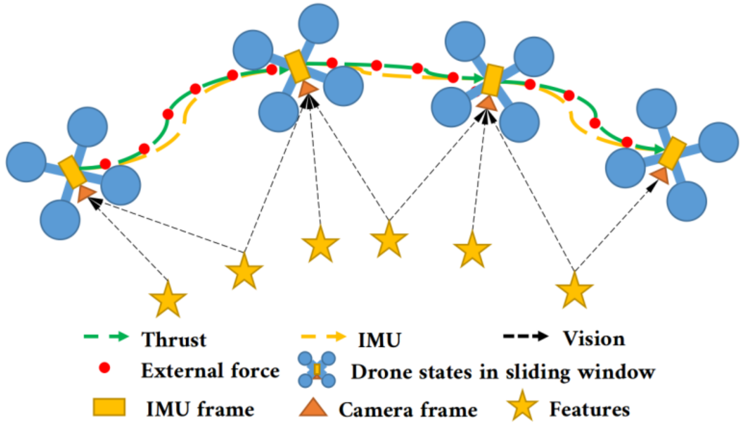
- Nonlinear System Identification
  - Inertial parameters
  - Geometric parameters
  - Dynamics parameters
- Newton-Euler Dynamics
  - Newton equation: $${}^{W}a = \left(\frac{1}{m}\right)\mathbf{R}_B^W\left(k_t u_i^2\, {}^B\mathbf{e}_3 - u_i D\, {}^B\mathbf{v}_i\right) - g\, {}^W\mathbf{e}_3$$
  - Euler equation: $$\begin{aligned}\mathbf{J}\, {}^B\alpha = &\sum_{i=1}^{n}\left(k_t u_i^2\, {}^B\mathbf{t}_B^i \times {}^B\mathbf{e}_3 - u_i\, {}^B\mathbf{t}_B^i \times D\, {}^B\mathbf{v}_i\right)\\ &+\sum_{i=1}^{n}\left(k_f u_i\, {}^B\mathbf{v}_i \times {}^B\mathbf{e}_3\right)\\ &+\sum_{i=1}^{n}\left((-1)^i k_m u_i^2\, {}^B\, {}^B\mathbf{e}_3\right)\\ &+\sum_{i=1}^{n}\left((-1)^{i+1} J_R u_i\, {}^B\omega \times {}^B\mathbf{e}_3\right)\\ &-{}^B\omega \times \mathbf{J}\, {}^B\omega\end{aligned}$$
- UKF Based Online Identification

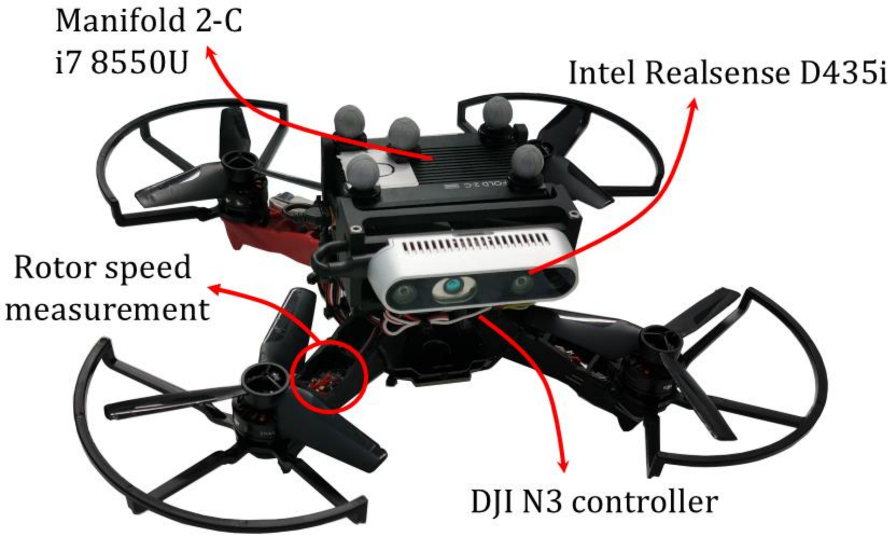


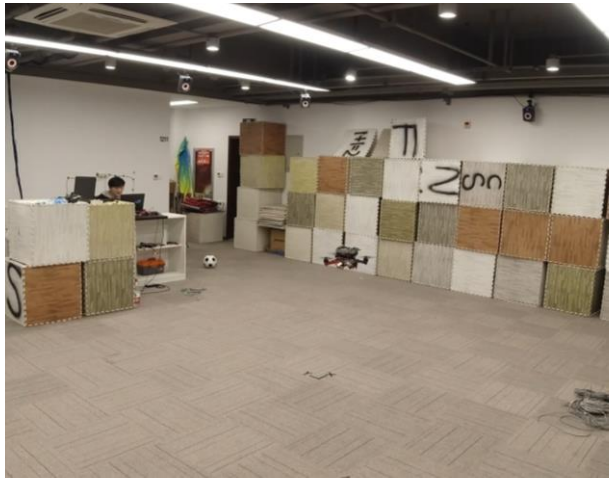
Vimo: Simultaneous visual inertial model-based odometry and force estimation, N. Barza, et al., IEEE Robotics and Automation Letters 2019
Vid-fusion: Robust visual-inertial-dynamics odometry for accurate external force estimation, Z. Ding, et al., IEEE ICRA 2021

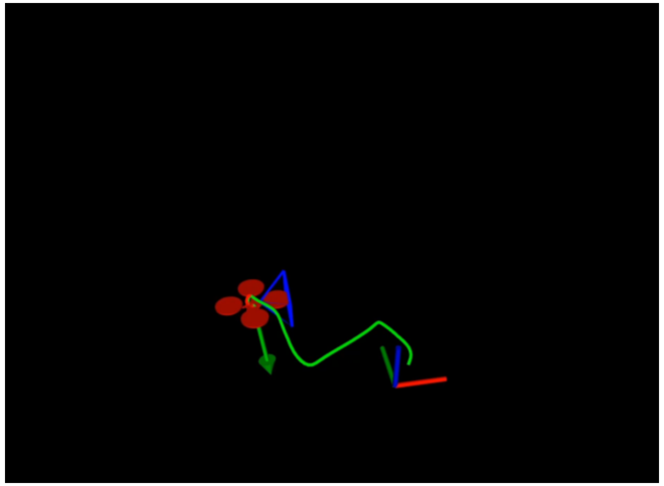
Disturbance-aware Navigation

➢ **Robust Visual-Inertial-Dynamics Odometry for Accurate External Force Estimation**









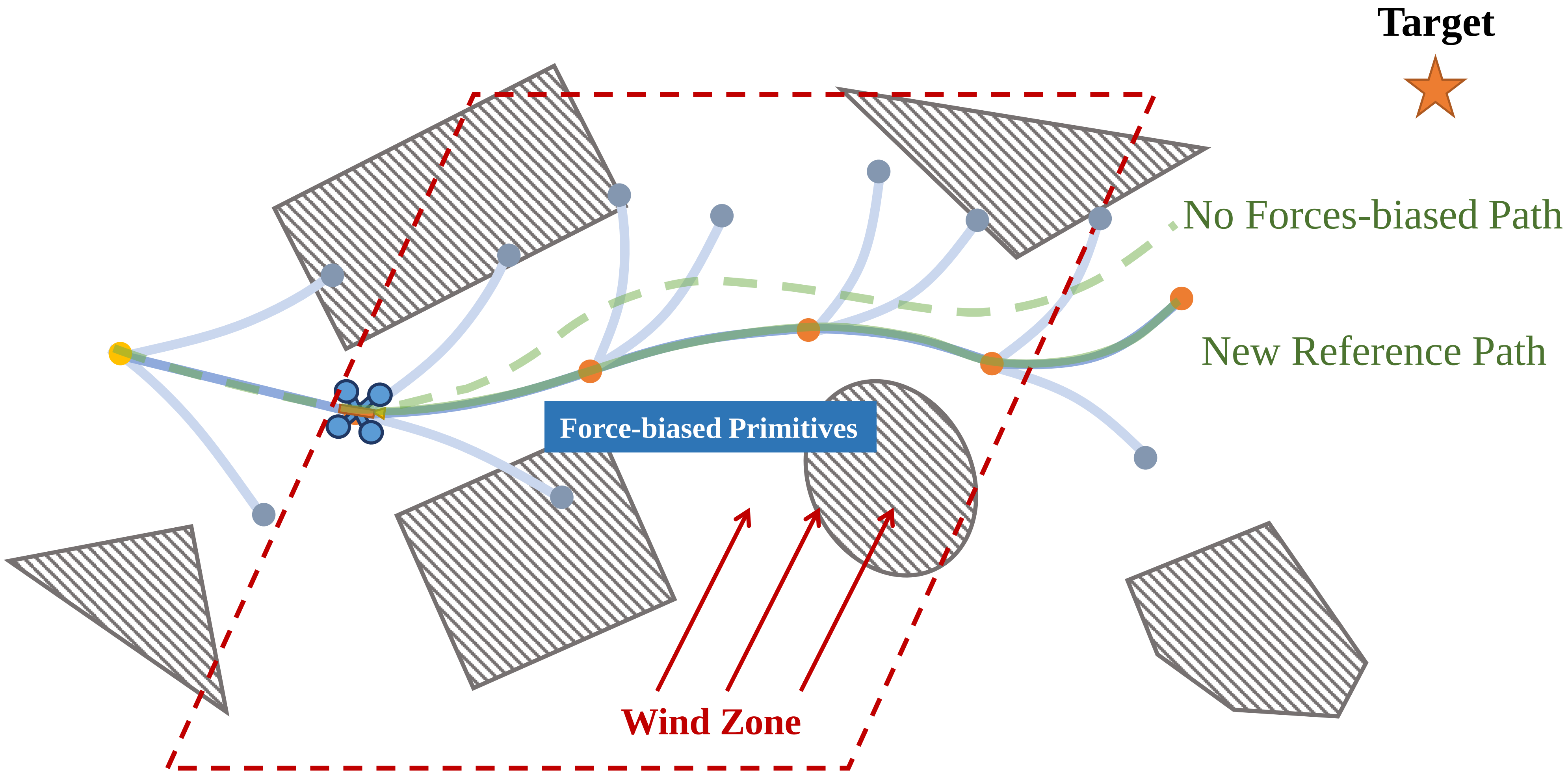
Flight with payload

Comparison with a force sensor

Vid-fusion: Robust visual-inertial-dynamics odometry for accurate external force estimation, Z. Ding, et al., IEEE ICRA 2021

Disturbance-aware Navigation

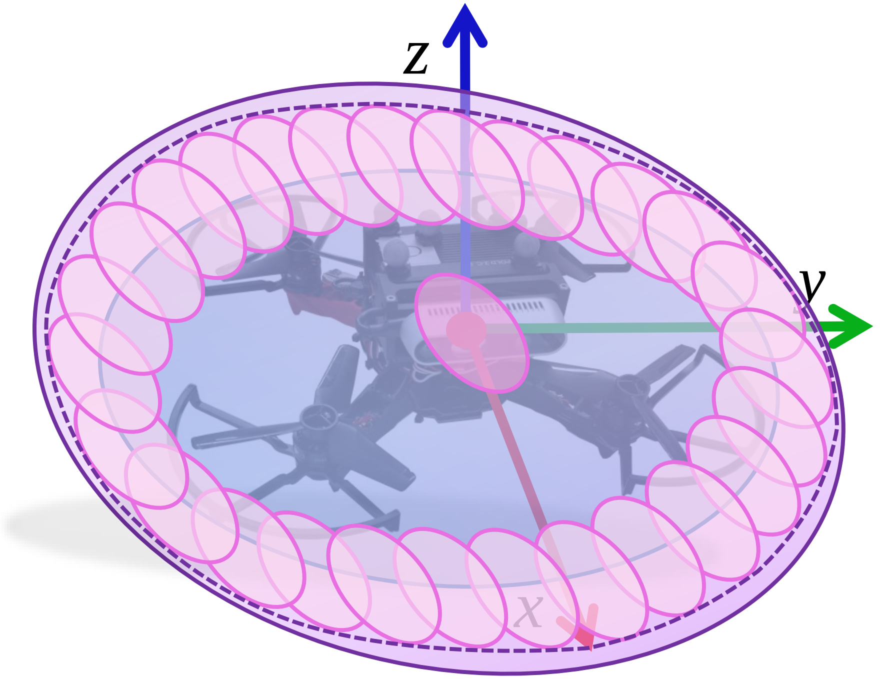
(1) Path Searching with Nominal Force



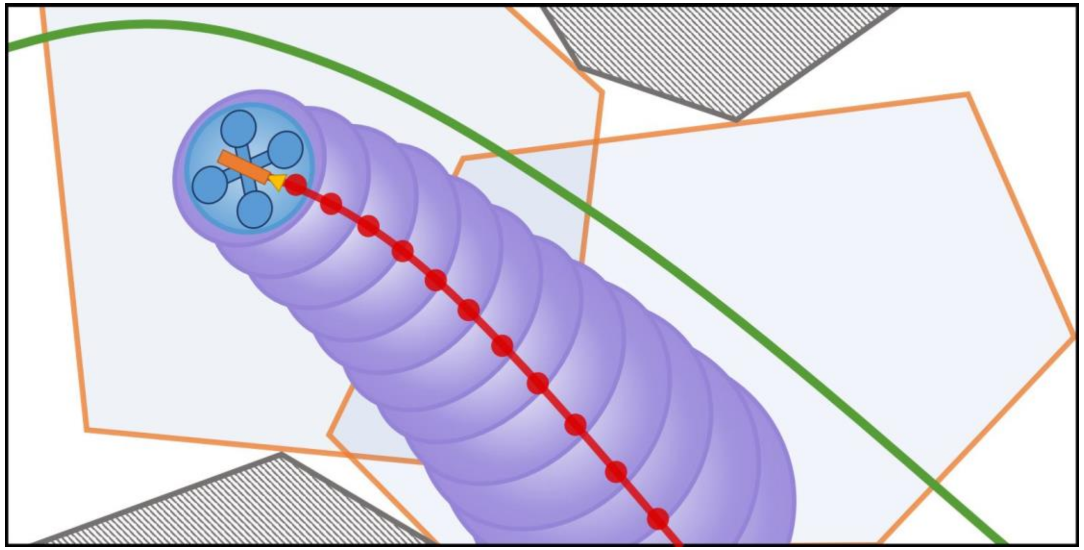
Disturbance-aware Navigation

(2) External Forces Resilient NMPC

Safe Corridor Constraints



- Ellipsoid of geometrical body
- Ellipsoid parameterizations of FRS[2]
- Outer ellipsoidal approximation
- Propagation of Safe Ellipsoid Boundary

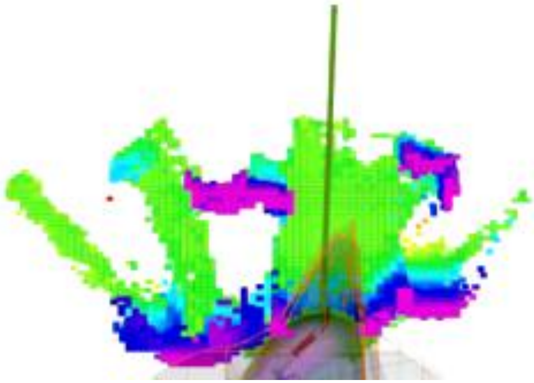


H. Seo, et al., Robust trajectory planning for a multirotor against disturbance based on hamilton-jacobi reachability analysis, IEEE/RSJ IROS 2019.

Disturbance-aware Navigation

(1) Indoor Flights Through Wind Zones (Proposed)

Drone Max Speed: 2.0 m/s
Wind Max Speed: 7.4 m/s



External Forces Resilient Safe Motion Planning for Quadrotor, Y. Wu, et al., IEEE Robotics and Automation Letters 2021

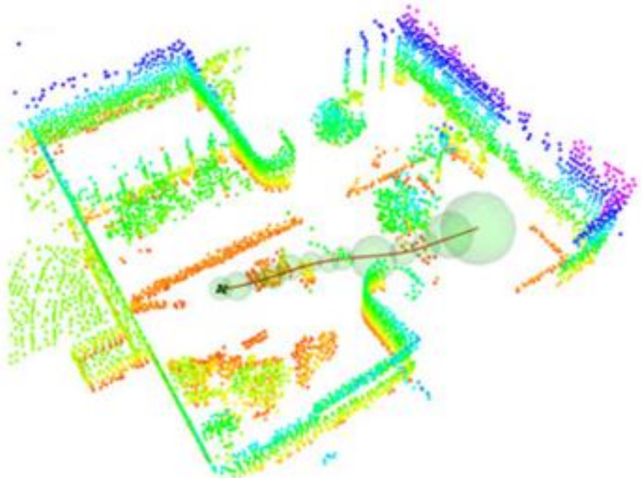
Agile: Optimal SE(3) Trajectory Planning

The Classical Formulation

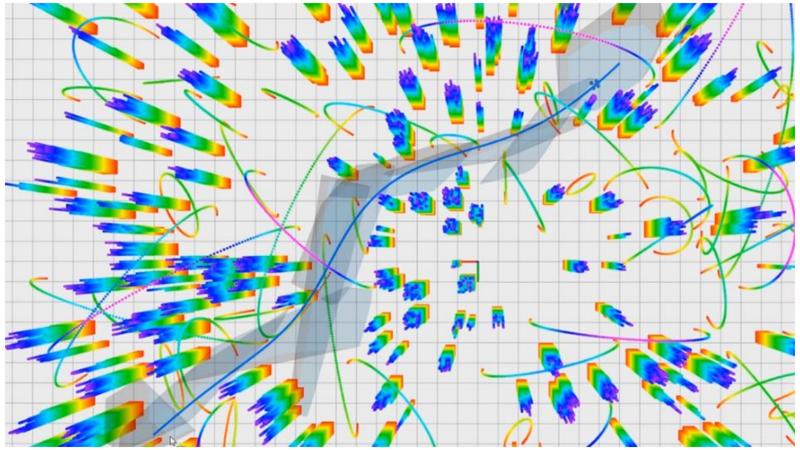
$$\rho_s(T) = \sum_{i=0}^{M_T} b_i T^i \quad \textbf{or} \quad \rho_f(T) = \begin{cases} 0 & \textit{if } T = T_\Sigma, \\ \infty & \textit{if } T \neq T_\Sigma. \end{cases}$$

$$\begin{aligned} \min_{z(t),T} \ & \int_0^T v(t)^{\mathrm{T}} \mathbf{W} v(t) \mathrm{d}t + \boxed{\rho(T)}, \\ s.t. \ & v(t) = z^{(s)}(t), \ \forall t \in [0, T], \\ & \mathcal{G}(z(t), \ldots, z^{(s)}(t)) \preceq \mathbf{0}, \ \forall t \in [0, T], \\ & \boxed{z(t) \in \mathcal{F}, \ \forall t \in [0, T],} \\ & z^{[s-1]}(0) = \bar{z}_o, \ z^{[s-1]}(T) = \bar{z}_f, \\ & z^{[s-1]} = (z^{\mathrm{T}}, \dot{z}^{\mathrm{T}}, \ldots, z^{(s-1)^{\mathrm{T}}})^{\mathrm{T}} \end{aligned}$$

Sphere SFC



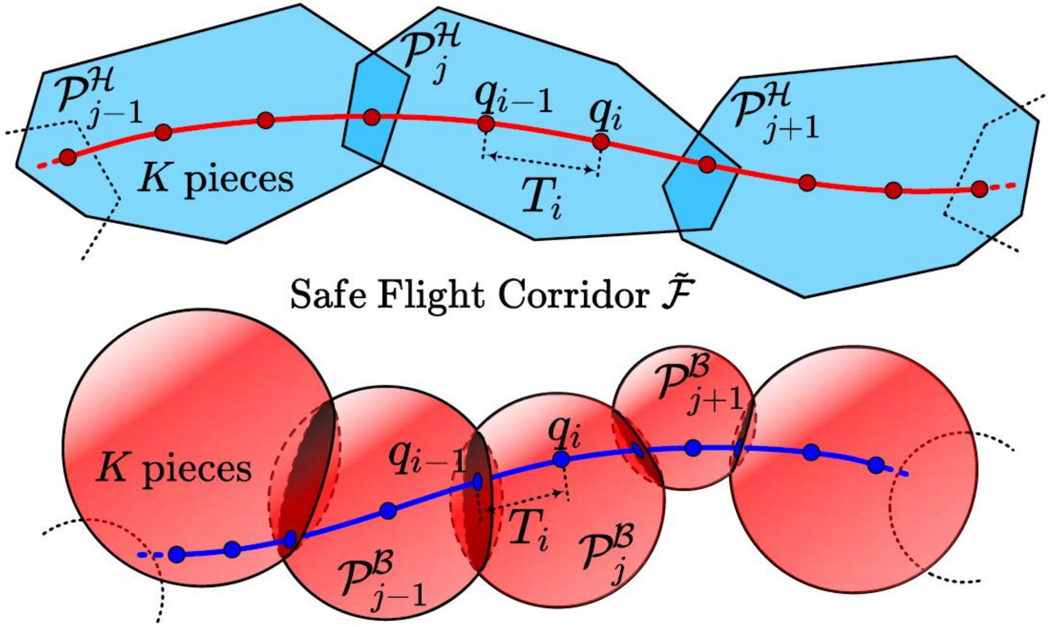
Polygon SFC



$\mathcal{F}$: SFC
(Safe Flight Corridor) Geometric Constraints

Optimal SE(3) Trajectory Planning

$$
\begin{aligned}
\min_{z(t),T} \; & \int_0^T v(t)^{\mathrm{T}} \mathbf{W} v(t) \mathrm{d}t + \boxed{\rho(T)} \\
s.t. \; & v(t) = z^{(s)}(t), \; \forall t \in [0, T], \\
& \mathcal{G}(z(t), \ldots, z^{(s)}(t)) \preceq \mathbf{0}, \; \forall t \in [0, T], \\
& z(t) \in \mathcal{F}, \; \forall t \in [0, T], \\
& z^{[s-1]}(0) = \bar{z}_o, \; z^{[s-1]}(T) = \bar{z}_f, \\
& z^{[s-1]} = (z^{\mathrm{T}}, \dot{z}^{\mathrm{T}}, \ldots, z^{(s-1)^{\mathrm{T}}})^{\mathrm{T}}
\end{aligned}
$$

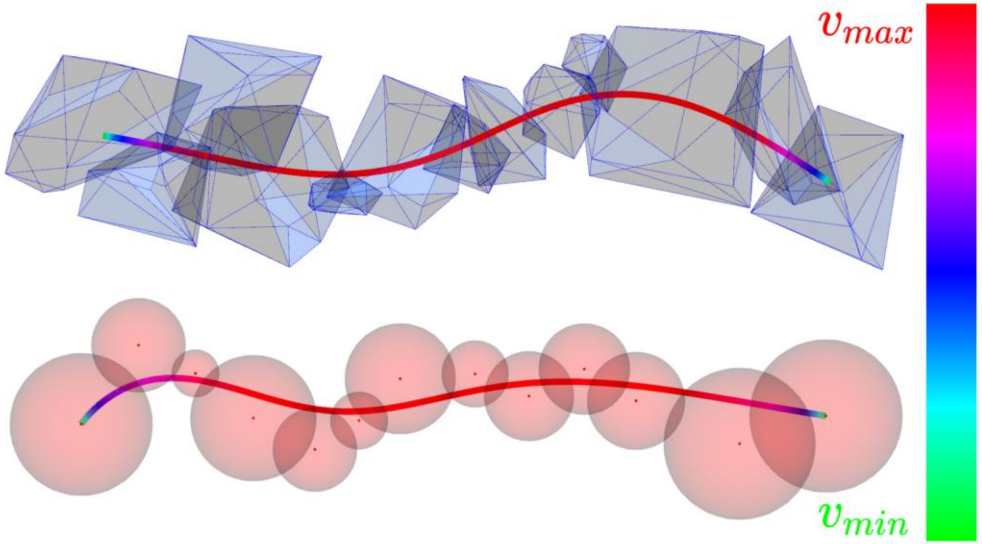


$$
\rho_f(T) = \begin{cases} 0 & \textit{if } T = T_\Sigma, \\ \infty & \textit{if } T \neq T_\Sigma. \end{cases}
$$

Fixed Total Time

$$
\rho_s(T) = \sum_{i=0}^{M_T} b_i T^i
$$

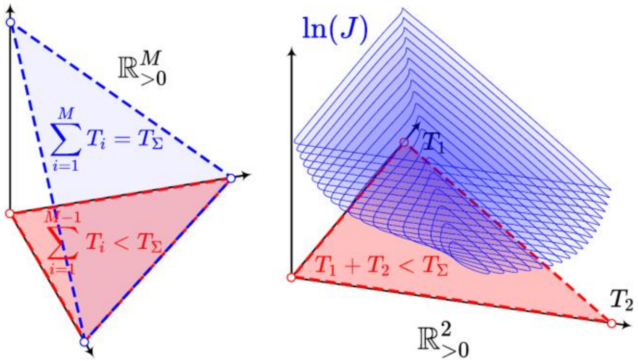
Time Regularization



Optimal SE(3) Trajectory Planning

Constraint Elimination

1. Fixed Total Time



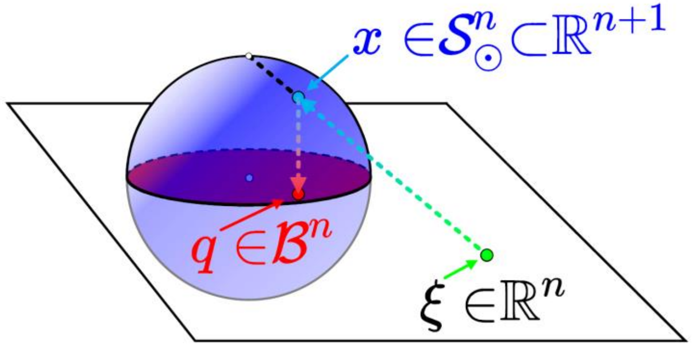
A simplex-like manifold, eliminated by the following diffeomorphism:

$$T_i = \frac{e^{\tau_i}}{1 + \sum_{j=1}^{M-1} e^{\tau_j}} T_\Sigma,$$

$$T_M = T_\Sigma - \sum_{i=1}^{M-1} T_i.$$

Now τ is in $\mathbb{R}^M$.

2. Ball-Shaped Free Space

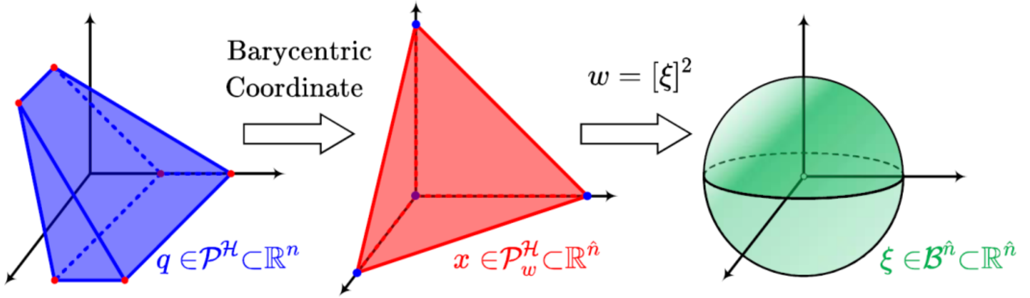


Lifted to n+1 dimension.
A diffeomorphism between n-sphere and $\mathbb{R}^n$ is used.

$$f_{\mathcal{B}}(x) = o + \frac{2r \cdot x}{x^{\mathrm{T}} x + 1}$$

Now x is in $\mathbb{R}^n$.

3. Polyhedron-Shaped Free Space



Polyhedron is also a ball after some smooth transformations.

$$f_{\mathcal{H}}(x) = v_0 + \frac{4\hat{\mathbf{V}}[x]^2}{(x^{\mathrm{T}} x + 1)^2}$$

Now x is in $\mathbb{R}^{\hat{n}}$.

First-order conditions for local minima are preserved by these smooth maps.

Optimal SE(3) Trajectory Planning

Trajectory Planning via Unconstrained NLP

$$\min_{z(t),T} \int_0^T v(t)^{\mathrm{T}} \mathbf{W} v(t) \mathrm{d}t + \rho(T),$$
$$s.t.\ \ v(t) = z^{(s)}(t),\ \forall t \in [0,T],$$
$$\boxed{\mathcal{G}(z(t),\ldots,z^{(s)}(t)) \preceq \mathbf{0},\ \forall t \in [0,T],}$$
$$z(t) \in \mathcal{F},\ \forall t \in [0,T],$$
$$z^{[s-1]}(0) = \bar{z}_o,\ z^{[s-1]}(T) = \bar{z}_f,$$
$$z^{[s-1]} = (z^{\mathrm{T}}, \dot{z}^{\mathrm{T}}, \ldots, z^{(s-1)^{\mathrm{T}}})^{\mathrm{T}}$$

Relaxation

$$\min_{\boldsymbol{\xi},\boldsymbol{\tau}} J(\mathbf{q}(\boldsymbol{\xi}), \mathbf{T}(\boldsymbol{\tau})) + I_{\Sigma}(\mathbf{c}(\mathbf{q}(\boldsymbol{\xi}), \mathbf{T}(\boldsymbol{\tau})), \mathbf{T}(\boldsymbol{\tau}))$$

Differential Flatness

$$\mathcal{G}_{\mathcal{D}}(x(t), u(t)) \preceq \mathbf{0},\ \forall t \in [0,T].$$

Differentiable Quadrature

$$I_{\mathcal{G}}^{k}[p] := \int_0^T \max\left[\mathcal{G}(p(t),\ldots,p^{(s)}(t)), \mathbf{0}\right]^k \mathrm{d}t$$

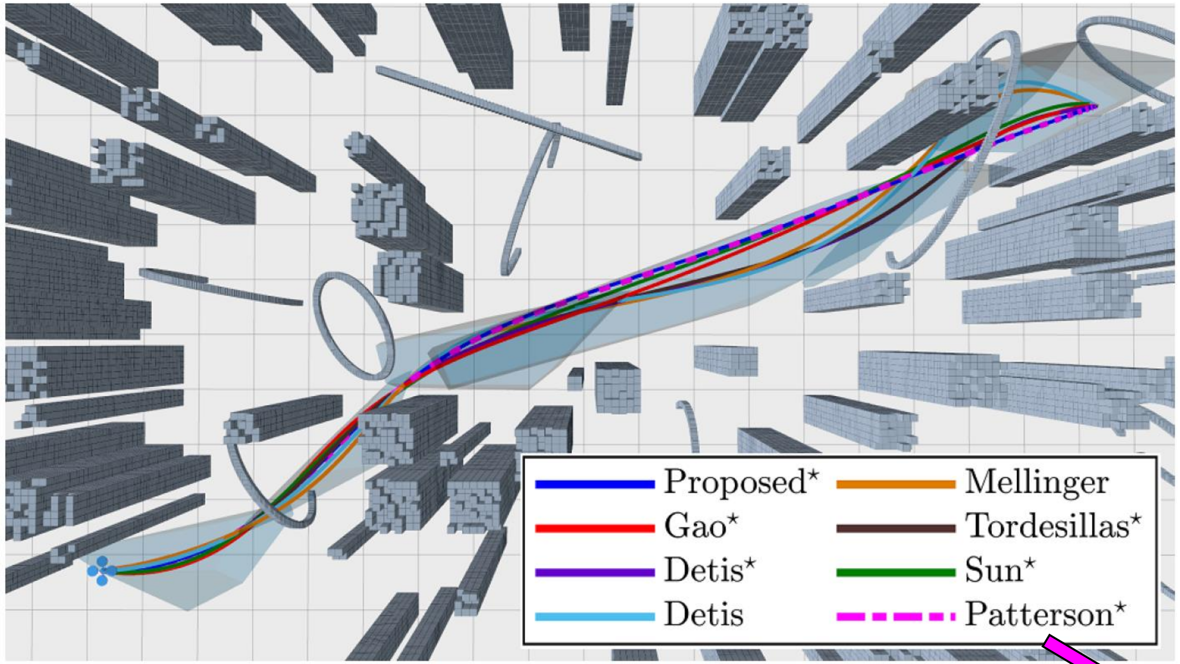
Features on the Relaxation:

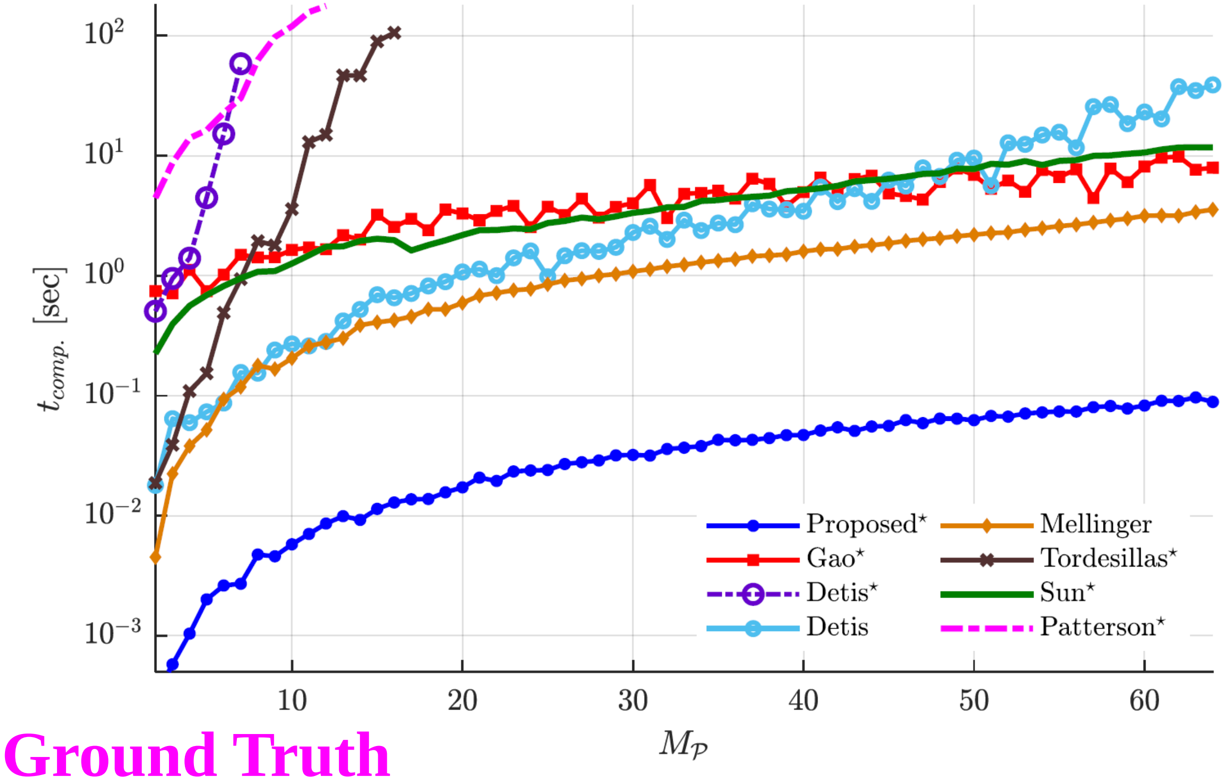
1. No Need for Initial Feasible Guesses;
2. Differentiating of Flatness Map Once and For All;
3. Handling Constraints on the True State and Control;
4. Unconstrained NLP is Fast.
5. Low-Dimensional Decision Variable.

……

Optimal SE(3) Trajectory Planning

Benchmarks with 7 classical or SOTA methods



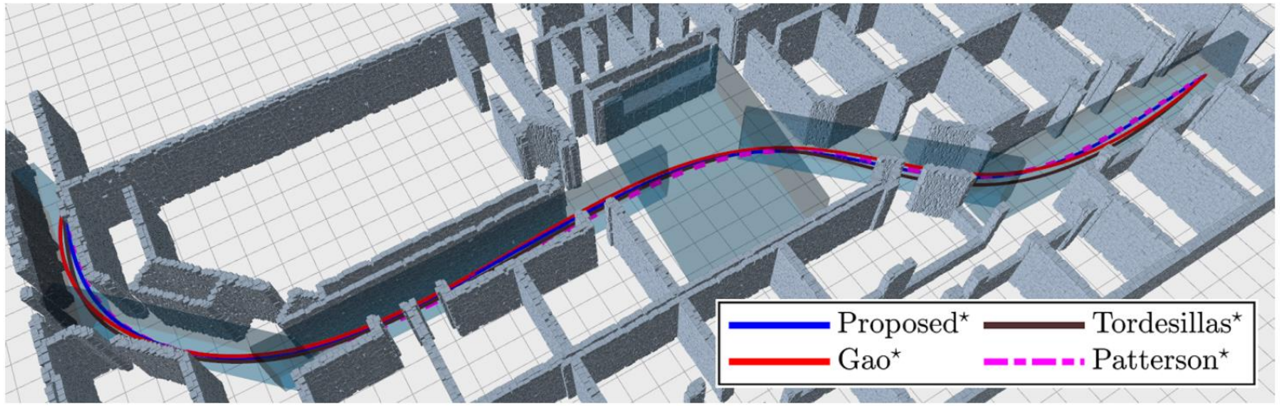


Solely C++11 and Eigen dependent, no commercial solvers.

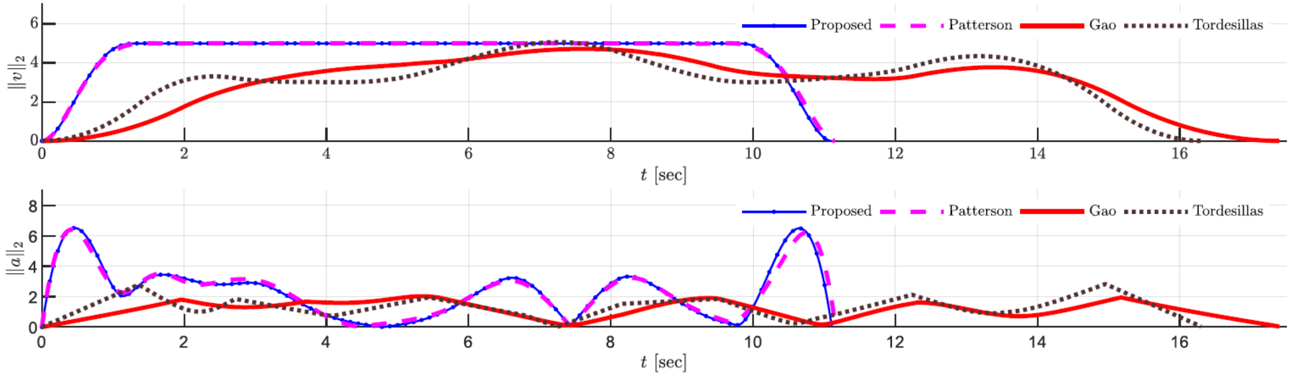
Ground Truth

Gao F, Wang L, Zhou B, Zhou X, Pan J and Shen S (2020) Teach-Repeat-Replan: A complete and robust system for aggressive flight in complex environments. IEEE Transactions on Robotics 36(5): 1526–1545.
Deits R and Tedrake R (2015b) Efficient mixed-integer planning for UAVs in cluttered environments. In: IEEE International Conference on Robotics and Automation. Seattle, USA, pp. 42–49.
Mellinger D and Kumar V (2011) Minimum snap trajectory generation and control for quadrotors. In: IEEE International Conference on Robotics and Automation. Shanghai, China, pp. 2520–2525.
Tordesillas J, Lopez BT and How JP (2019) FASTER: Fast and safe trajectory planner for flights in unknown environments. In: IEEE International Conference on Intelligent Robots and Systems. Macau, China, pp. 1934–1940.
Sun W, Tang G and Hauser K (2020) Fast UAV trajectory optimization using bilevel optimization with analytical gradients. In: American Control Conference. pp. 82–87.
Patterson MA and Rao AV (2014) GPOPS-II: A MATLAB software for solving multiple-phase optimal control problems using hpadaptive gaussian quadrature collocation methods and sparse nonlinear programming. ACM Transactions on Mathematical Software 41(1): 1–37.

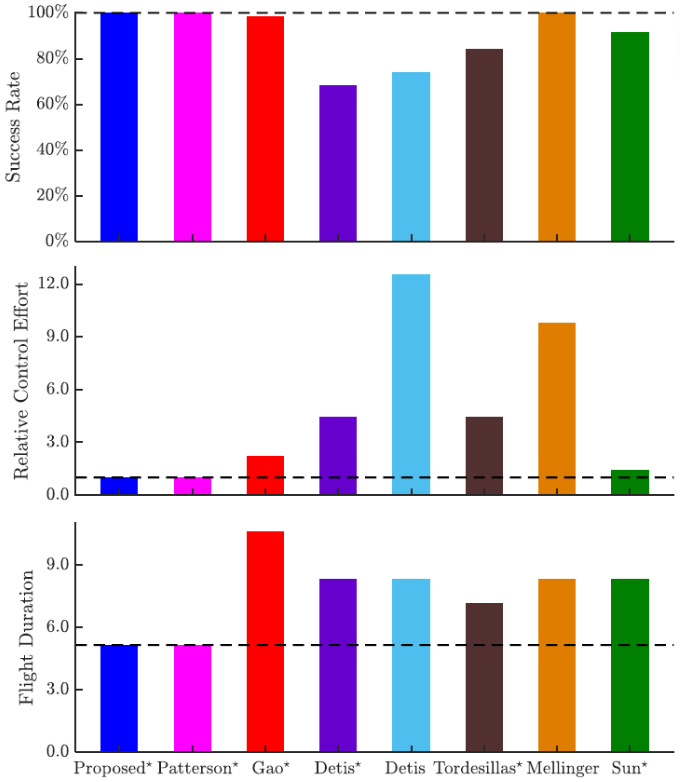
Optimal SE(3) Trajectory Planning

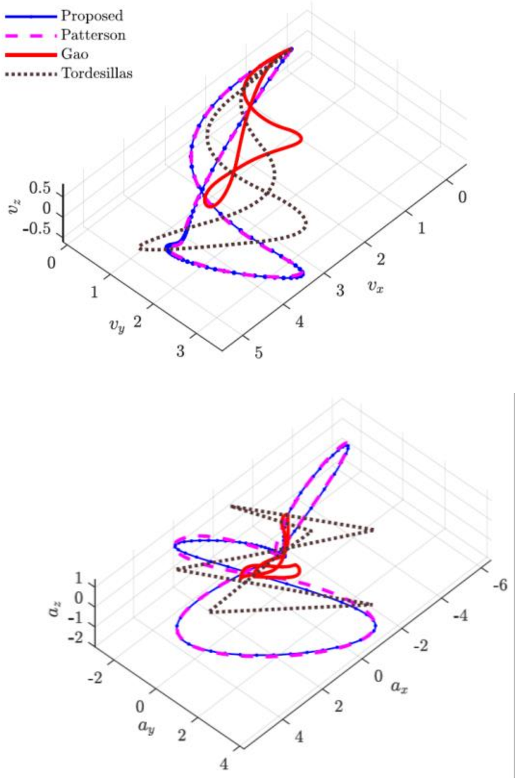


(a) Trajectories generated by different methods within a long SFC of the office environment



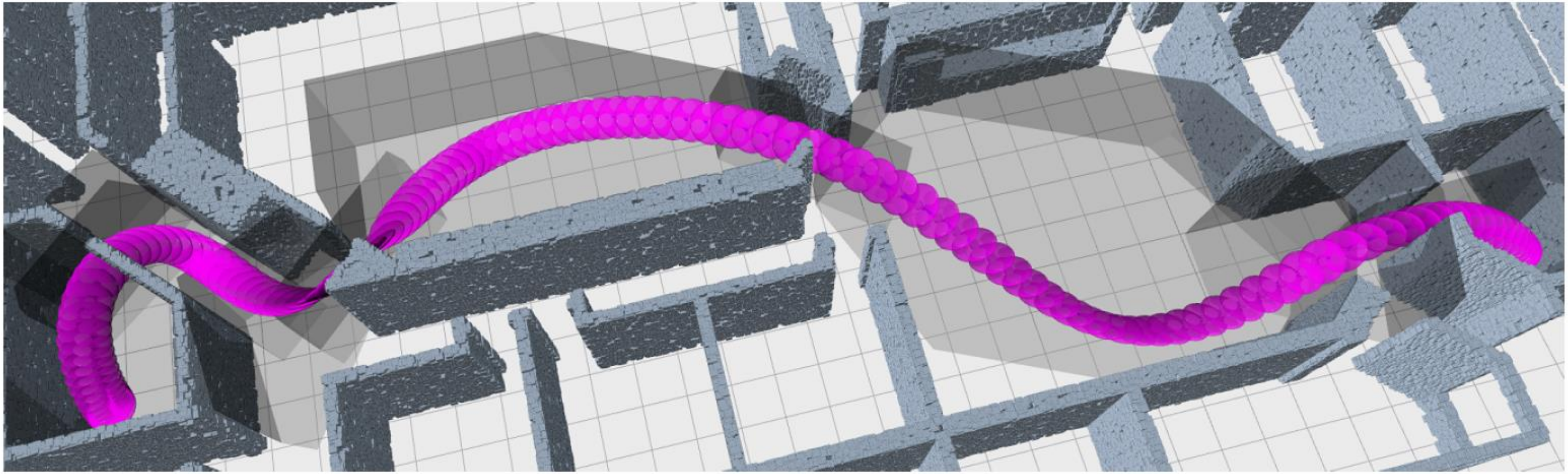
(b) The velocity and acceleration magnitude for different methods.





Gao F, Wang L, Zhou B, Zhou X, Pan J and Shen S (2020) Teach-Repeat-Replan: A complete and robust system for aggressive flight in complex environments. IEEE Transactions on Robotics 36(5): 1526–1545.
Deits R and Tedrake R (2015b) Efficient mixed-integer planning for UAVs in cluttered environments. In: IEEE International Conference on Robotics and Automation. Seattle, USA, pp. 42–49.
Mellinger D and Kumar V (2011) Minimum snap trajectory generation and control for quadrotors. In: IEEE International Conference on Robotics and Automation. Shanghai, China, pp. 2520–2525.
Tordesillas J, Lopez BT and How JP (2019) FASTER: Fast and safe trajectory planner for flights in unknown environments. In: IEEE International Conference on Intelligent Robots and Systems. Macau, China, pp. 1934–1940.
Sun W, Tang G and Hauser K (2020) Fast UAV trajectory optimization using bilevel optimization with analytical gradients. In: American Control Conference. pp. 82–87.
Patterson MA and Rao AV (2014) GPOPS-II: A MATLAB software for solving multiple-phase optimal control problems using hpadaptive gaussian quadrature collocation methods and sparse nonlinear programming. ACM Transactions on Mathematical Software 41(1): 1–37.

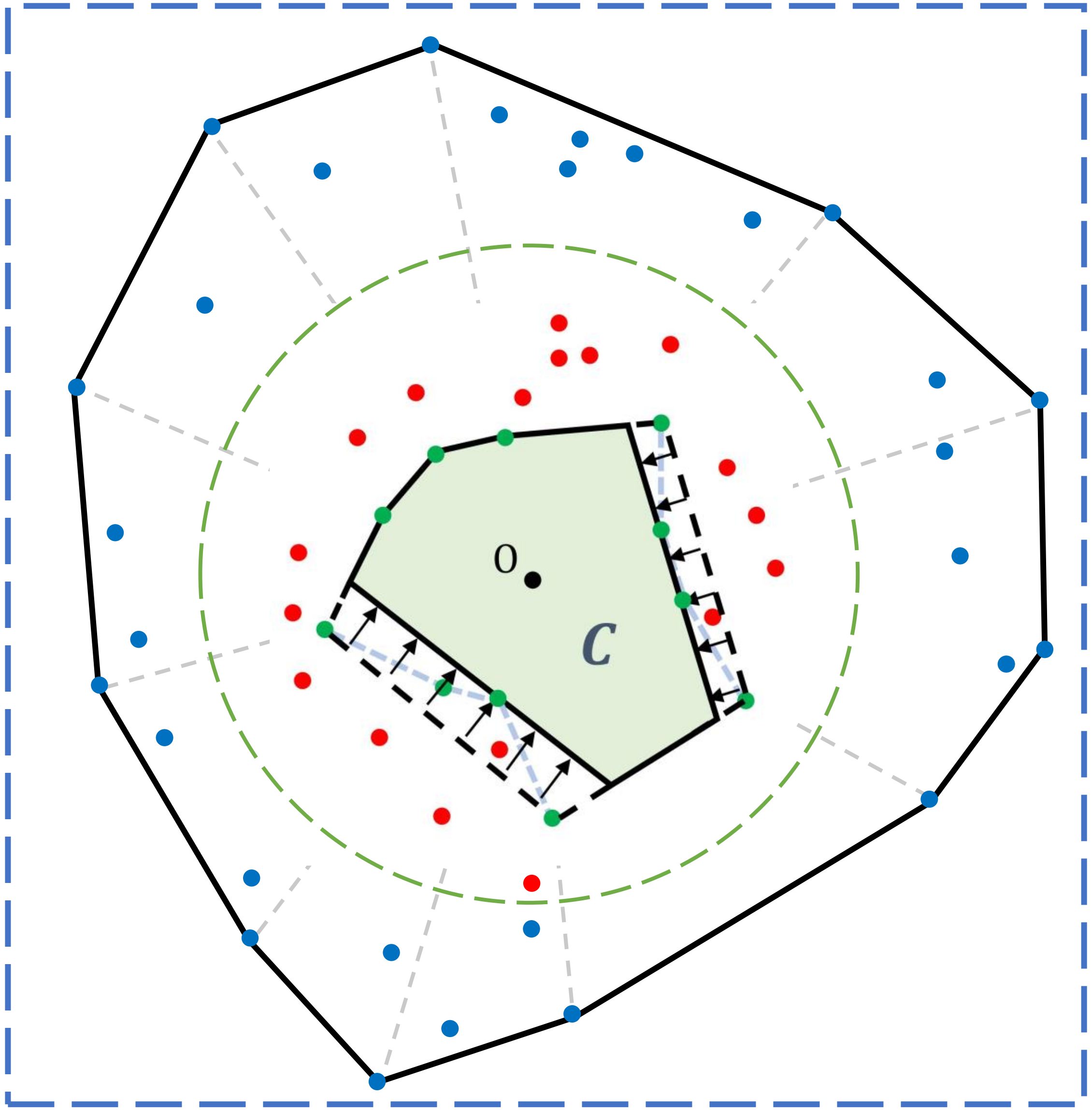
Optimal SE(3) Trajectory Planning



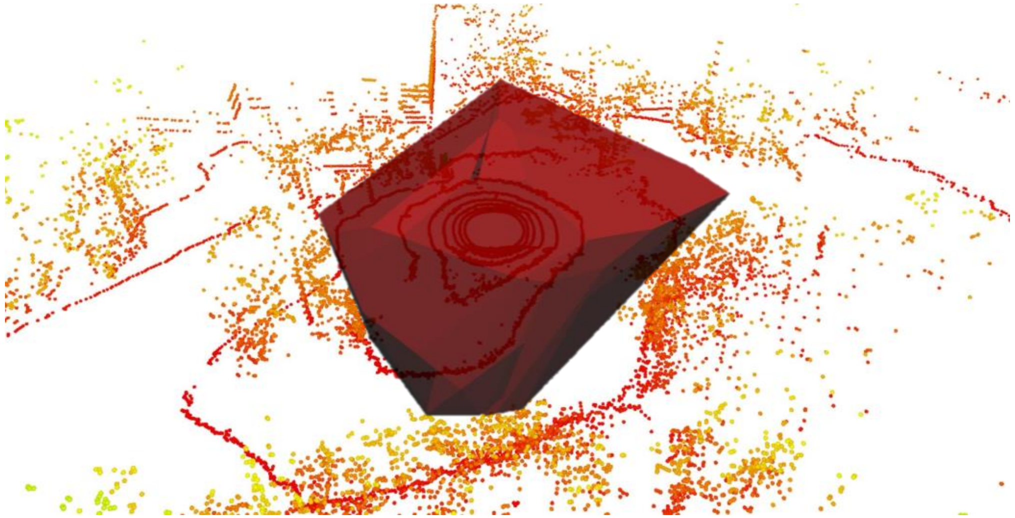
Computing time： 42.88ms

Geometrically Constrained Trajectory Optimization for Multicopters, Zhepei Wang, Fei Gao, et al., arXiv preprint arXiv:2103.00190

Optimal SE(3) Trajectory Planning



Methodology



Real world result

TEST ON LIDAR DATA SET

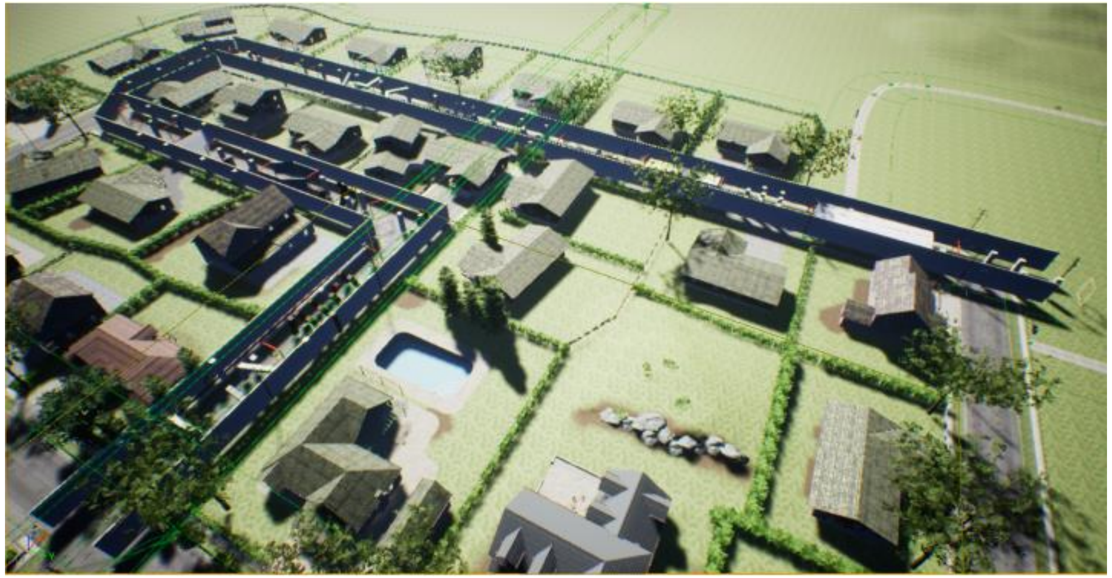
	non-iterative IRIS	IRIS	Ours
Average volume	37.3002	**645.107**	499.022
Average time(ms)	145.662	1279.66	**7.22958**

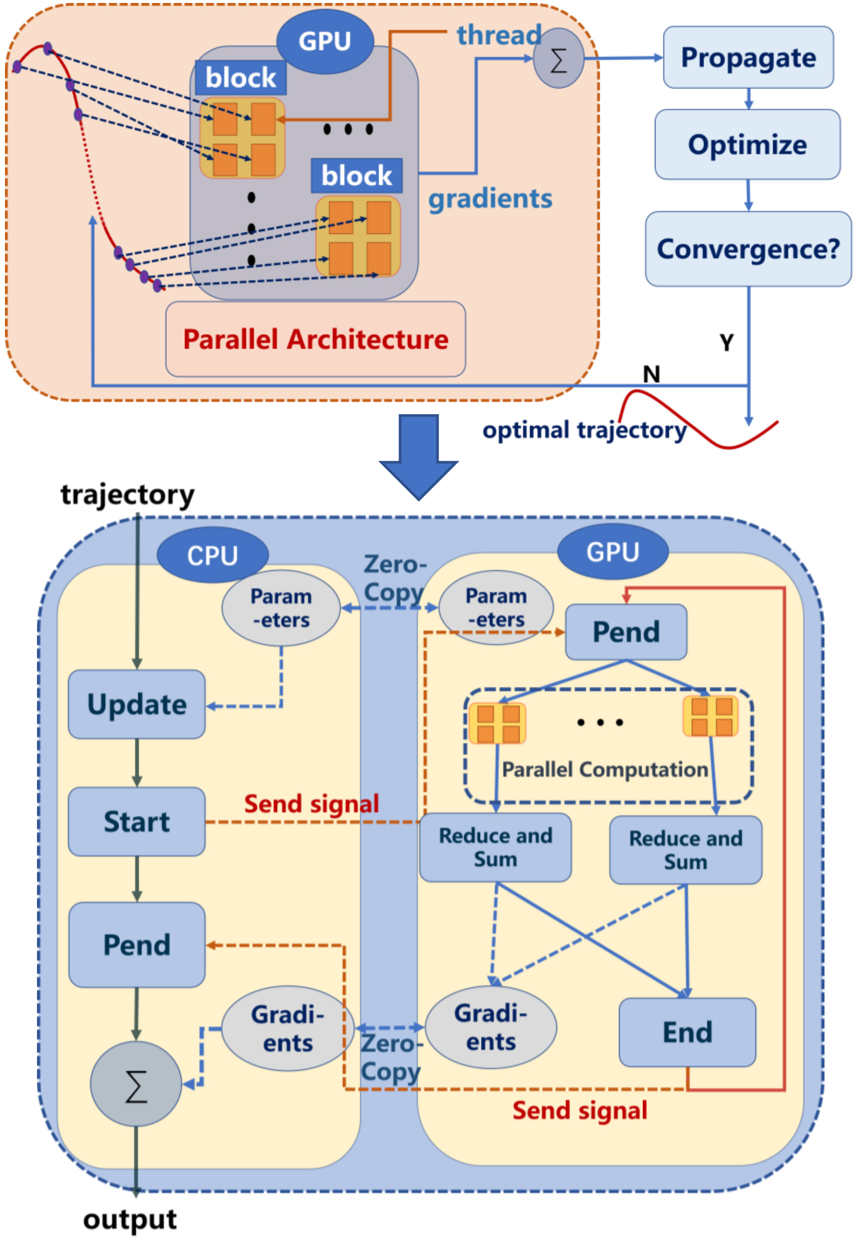
TEST ON RANDOM DATA

		non-iterative IRIS	IRIS	Ours
Sphere	Average volume	903.088	**1601.98**	1512.11
	Average time(ms)	42.184	825.732	**7.82334**
cuboid	Average volume	81.0409	**532.232**	350.213
	Average time(ms)	10.1022	426.489	**2.84751**
cross	Average volume	23.4778	**3018.49**	1720.2
	Average time(ms)	11.0267	304.786	**1.99532**

Much faster than SOTA

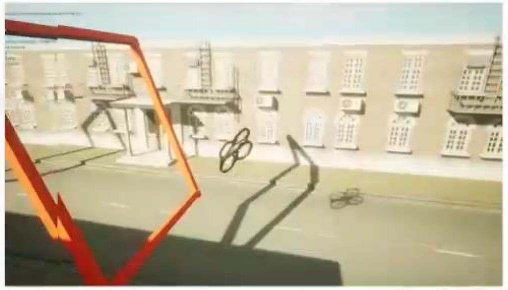
Optimal SE(3) Trajectory Planning

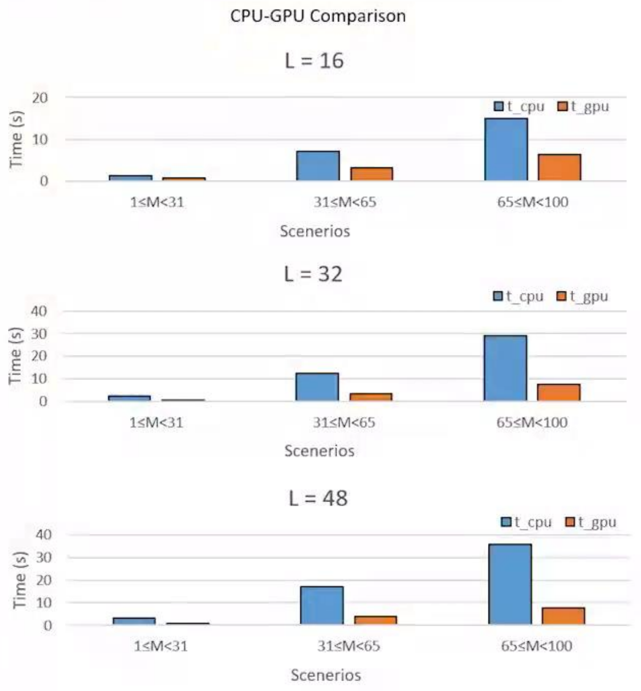




Optimal SE(3) Trajectory Planning

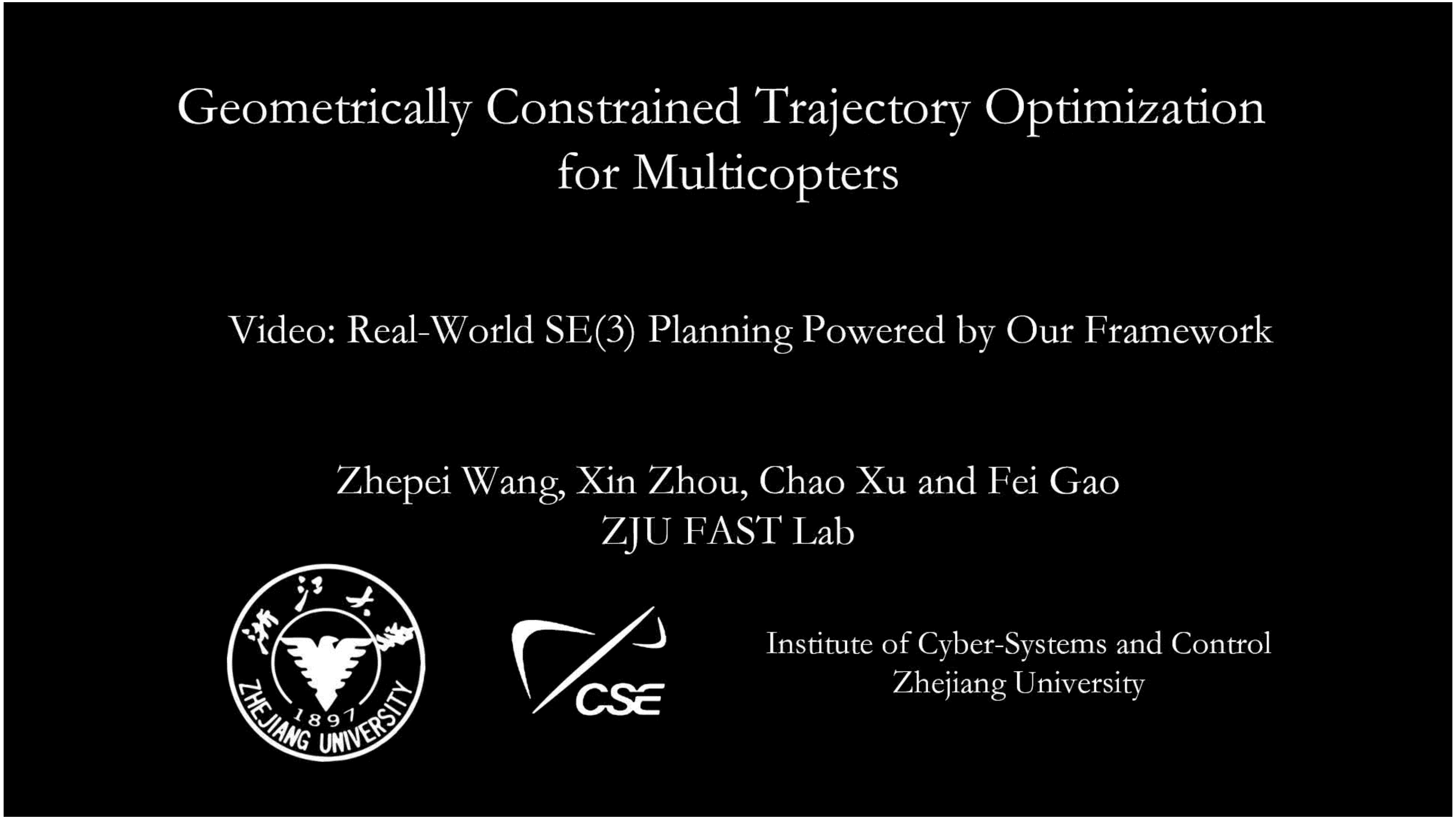
A video clip demonstrates that our drone aggressively and safely zooms between surrounding obstacles in a large attitude.





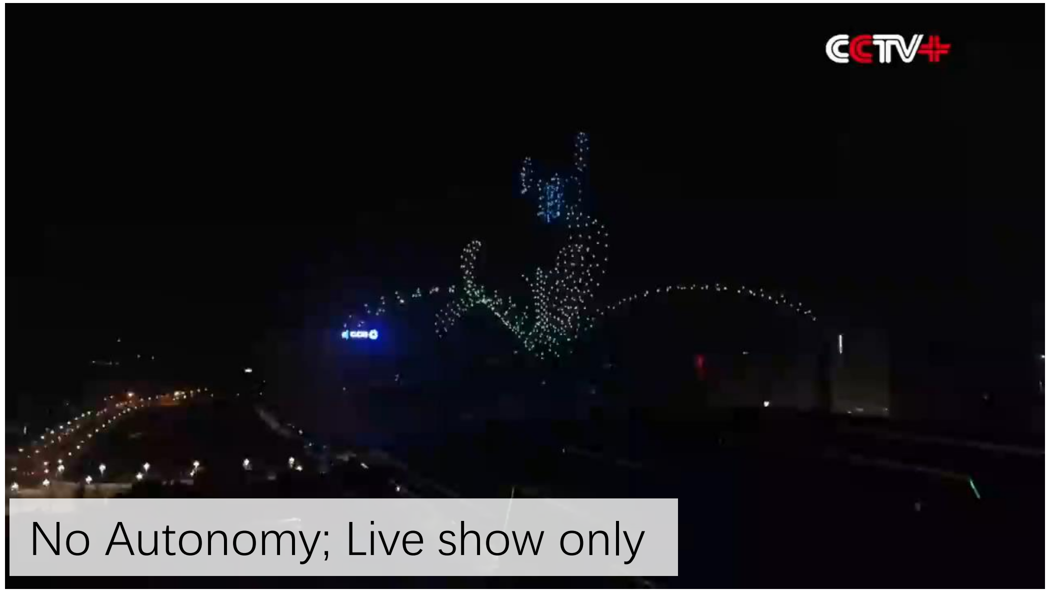
L is the number of trajectory segments
M is the number of constraint points on each trajectory segment.

Optimal SE(3) Trajectory Planning



Geometrically Constrained Trajectory Optimization for Multicopters, Zhepei Wang, Fei Gao, arXiv preprint arXiv:2103.00190.

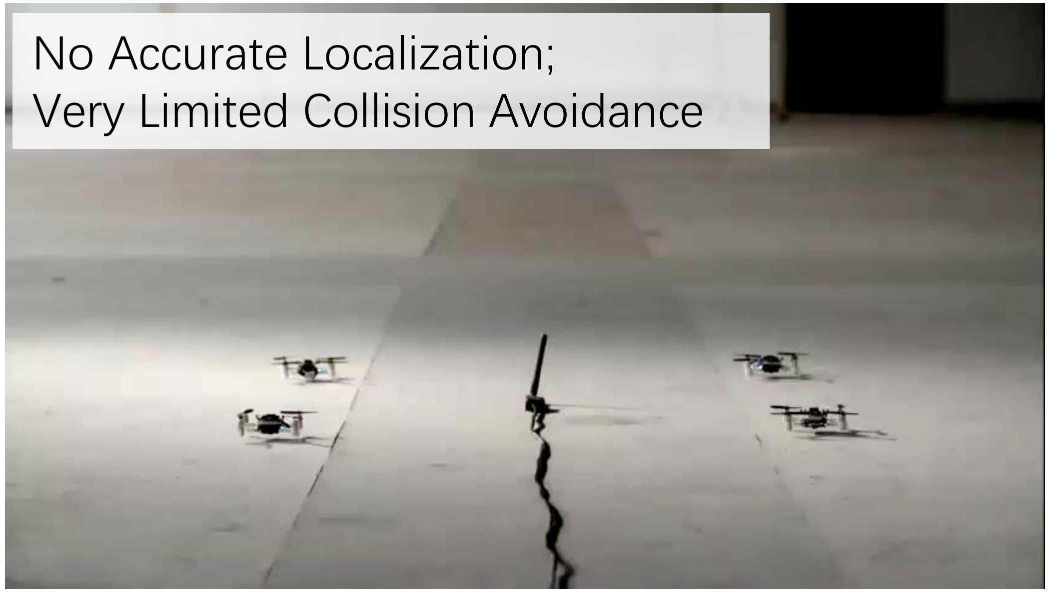
Swarm: Decentralized and Fully Autonomous Aerial Swarm



No Autonomy; Live show only

Motion Capture System Required

Centralized Planning;
No Collision Avoidance with Environments

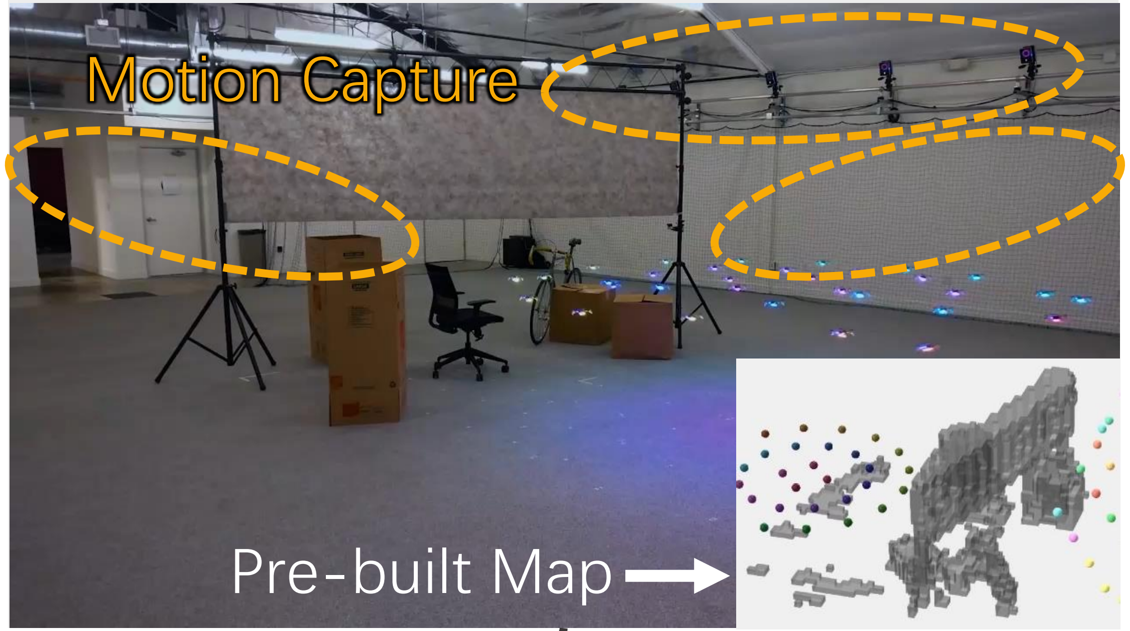


No Accurate Localization;
Very Limited Collision Avoidance

Decentralized and Fully Autonomous Aerial Swarm

Single-drone Autonomous Navigation

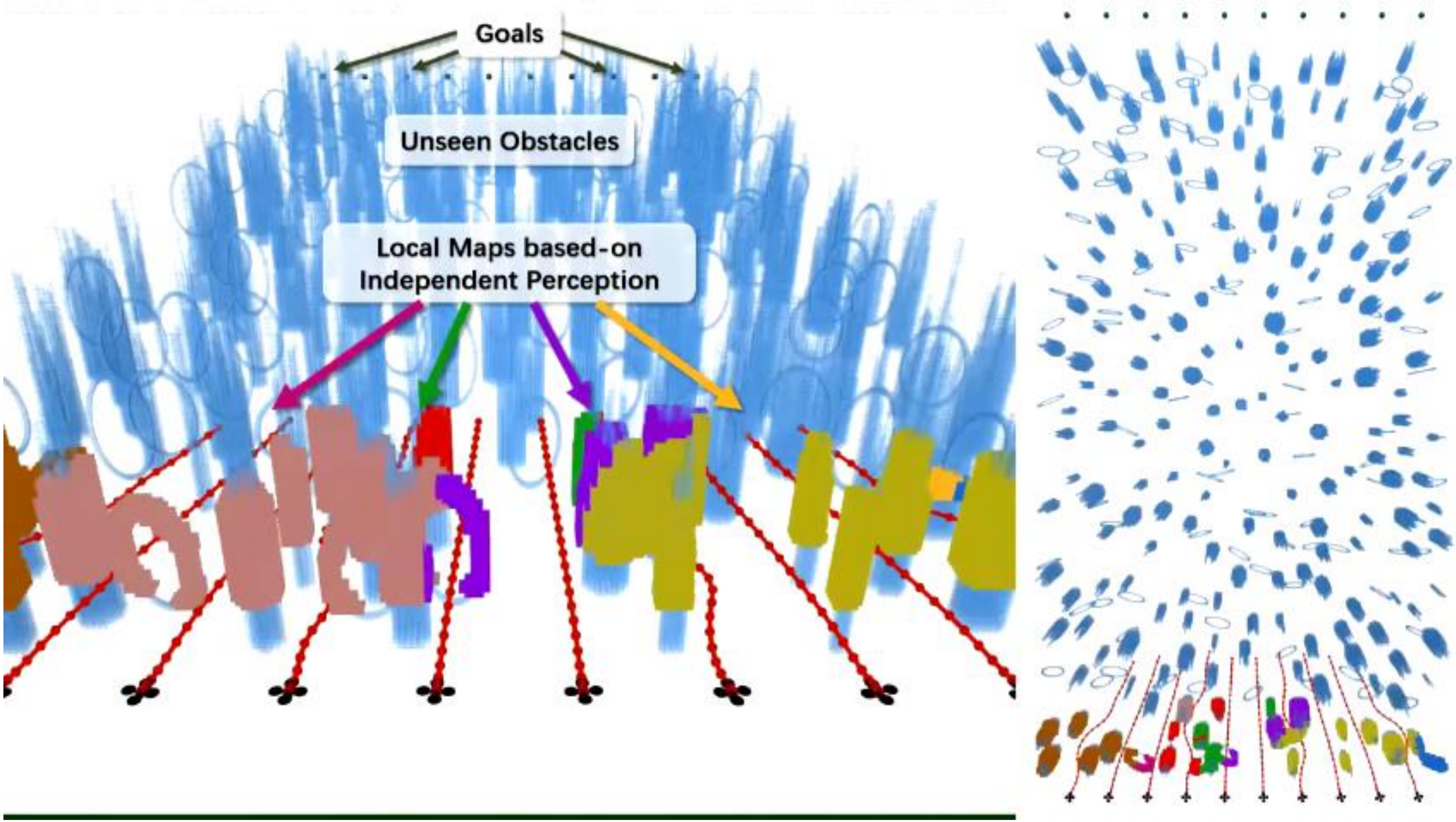
Offline Drone-swarm Planning



We propose a systematic solution for fully autonomous multi-drone navigation using merely onboard resources.

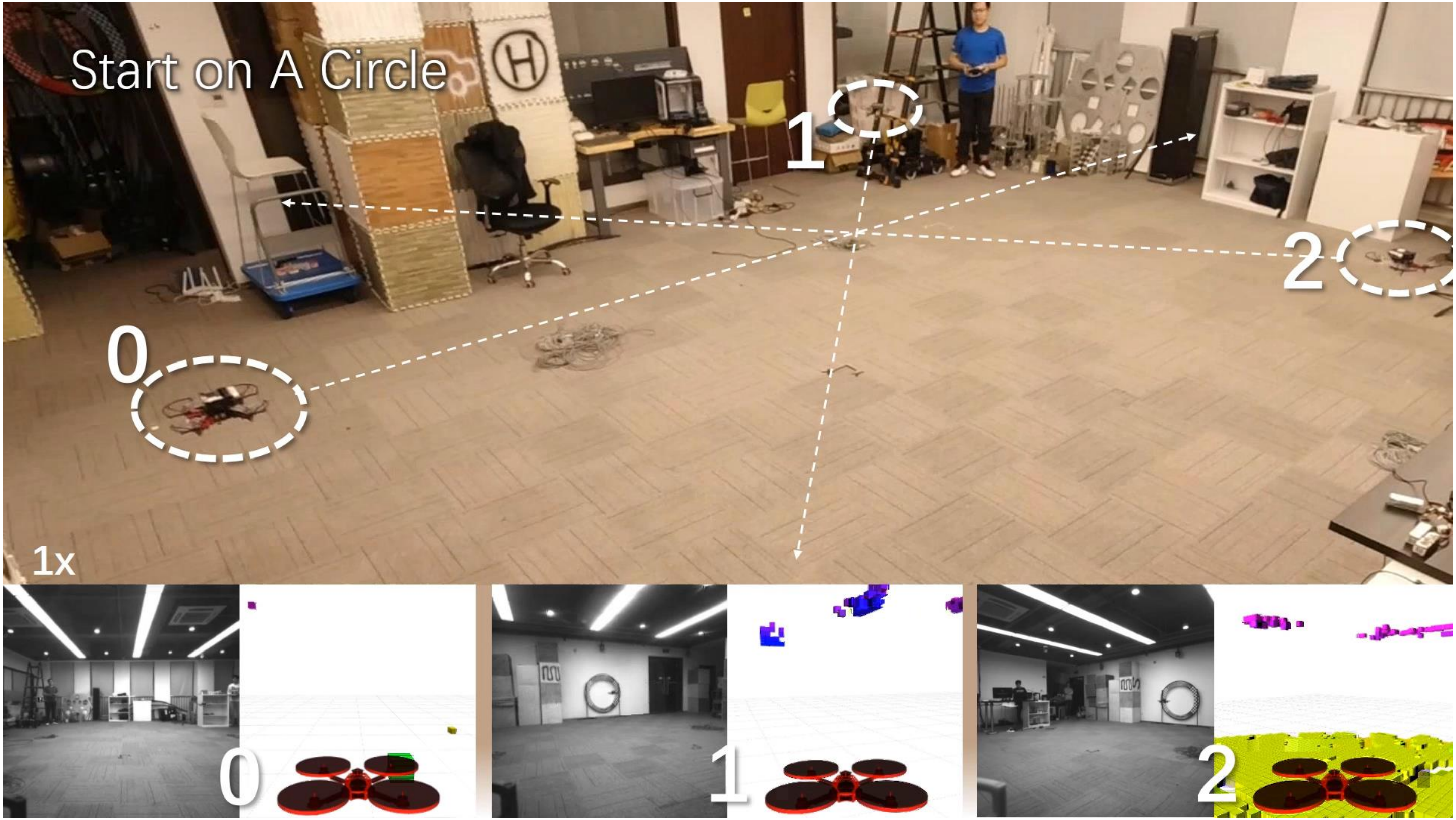
Left video: EGO-Planner: An ESDF-free Gradient-based Local Planner for Quadrotors, Xin Zhou, Fei Gao, et al., IEEE Robotics and Automation Letters 2021
Right video: Trajectory planning for quadrotor swarms, Hönig, Wolfgang, et al., IEEE Transactions on Robotics 2018

Decentralized and Fully Autonomous Aerial Swarm



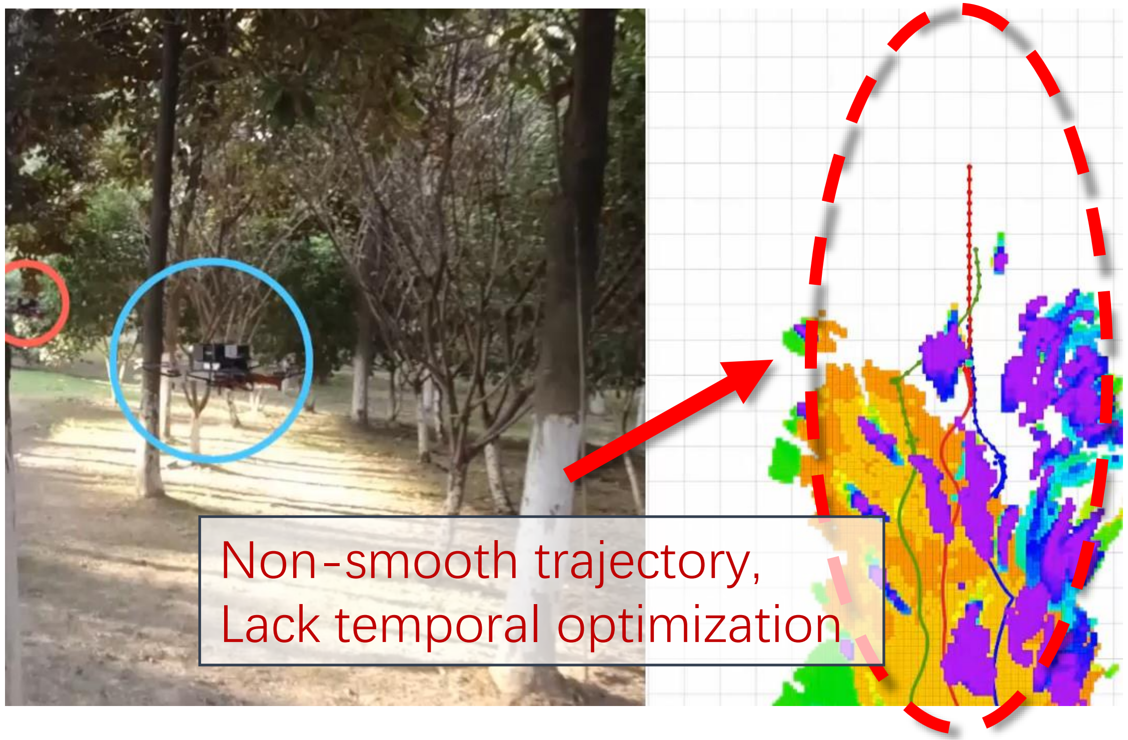
EGO-Swarm: A Fully Autonomous and Decentralized Quadrotor Swarm System in Cluttered Environments, Xin Zhou, Fei Gao, et al., IEEE ICRA 2021

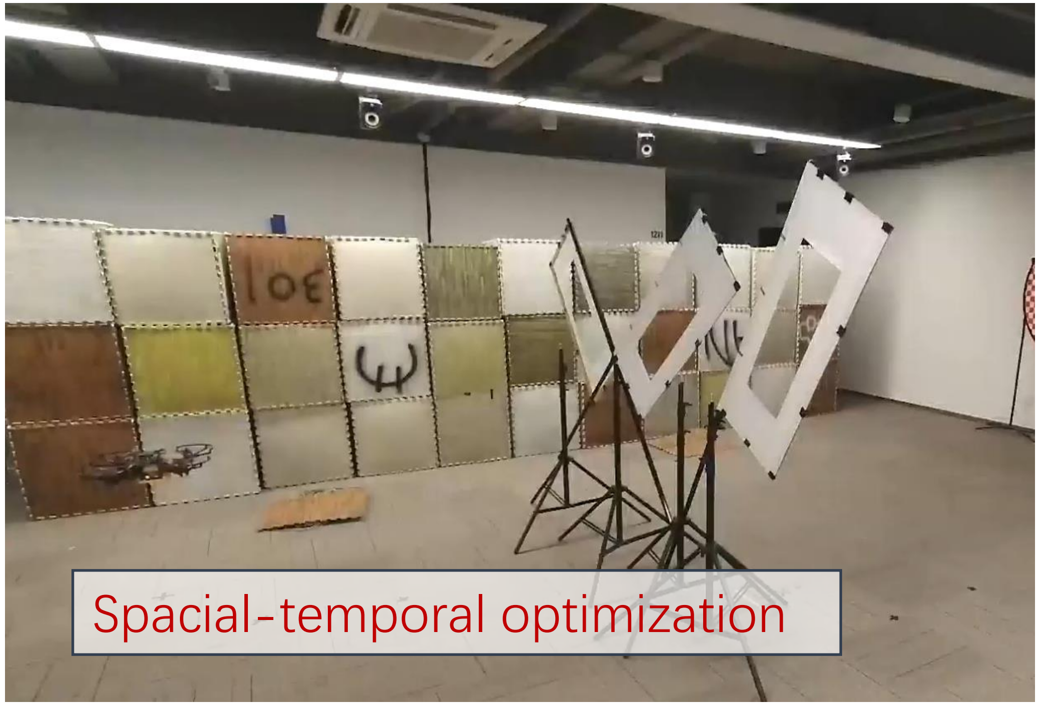
Decentralized and Fully Autonomous Aerial Swarm



EGO-Swarm: A Fully Autonomous and Decentralized Quadrotor Swarm System in Cluttered Environments, Xin Zhou, Fei Gao, et al., IEEE ICRA 2021

Decentralized and Fully Autonomous Aerial Swarm



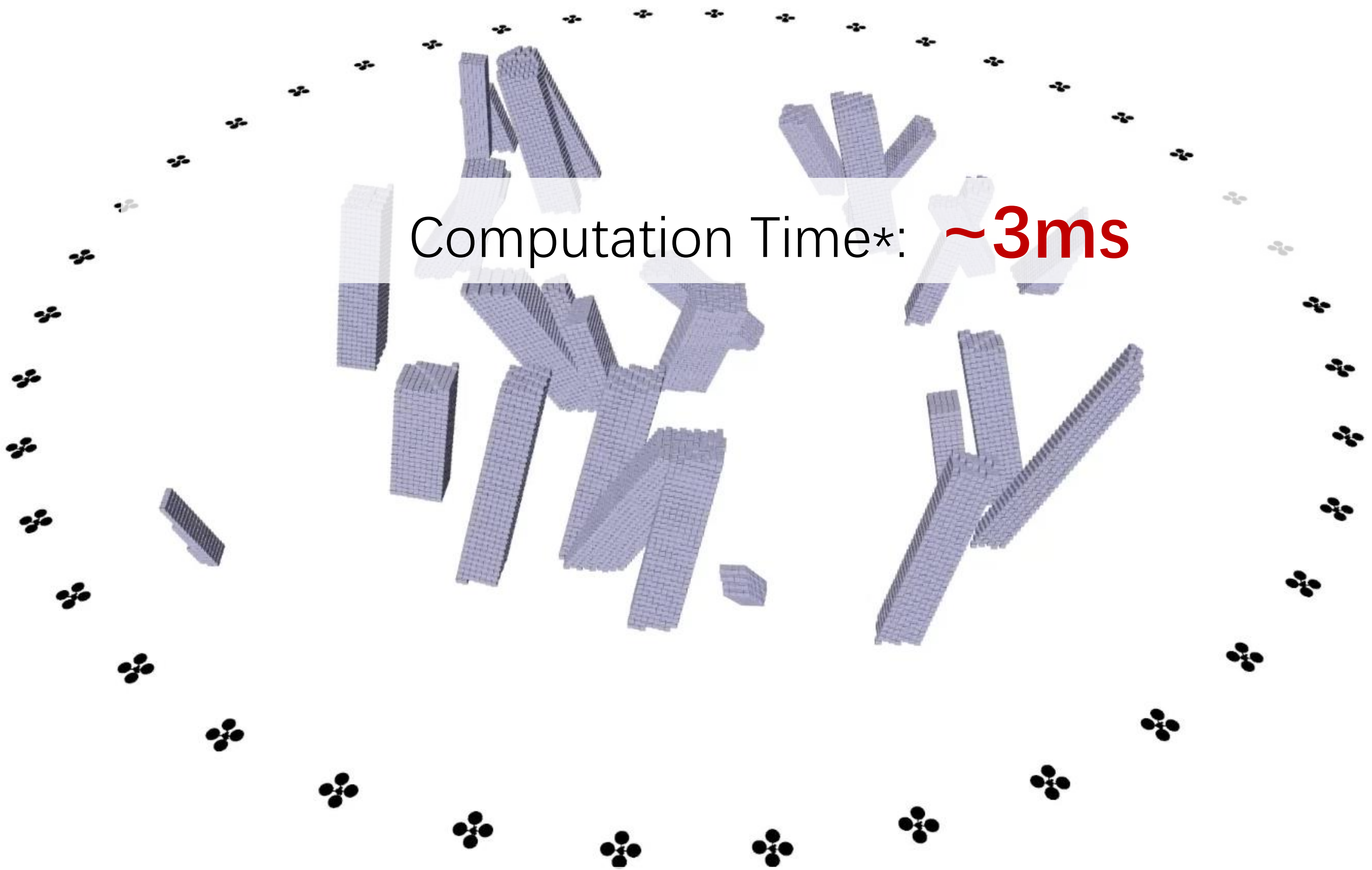


The proposed planner features

1. Decentralization and asynchronism
2. Spatial-temporal optimization
3. Real-world validation

Left: EGO-Swarm: A Fully Autonomous and Decentralized Quadrotor Swarm System in Cluttered Environments, Xin Zhou, Fei Gao, et al., IEEE ICRA 2021
Right: Geometrically Constrained Trajectory Optimization for Multicopters, Zhepei Wang, Fei Gao, et al., arXiv preprint arXiv:2103.00190

Decentralized and Fully Autonomous Aerial Swarm



* Decentralized system. Computation time of each replan of each agent. Tested on Intel Core i7 10700KF

Decentralized Spatial-Temporal Trajectory Planning for Multicopter Swarms, Xin Zhou, Fei Gao, et al., arXiv preprint arXiv: 2106.12481

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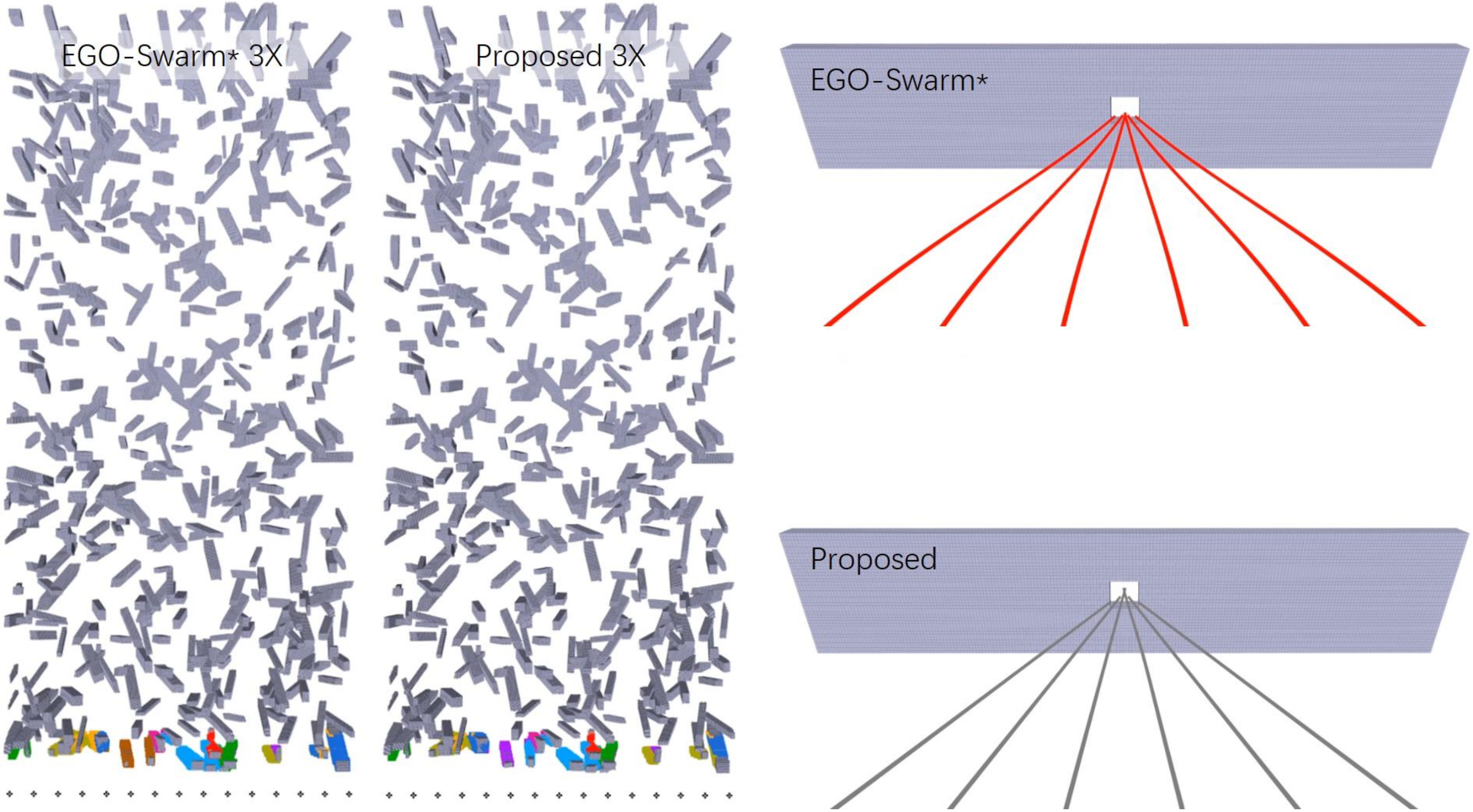

Decentralized Spatial-Temporal Trajectory Planning for Multicopter Swarms, Xin Zhou, Fei Gao, et al., arXiv preprint arXiv: 2106.12481

Thanks.